nDNA: Layerwise Prediction Profiles for Comparing Language-Model Variants

Anonymous ACL submission

Abstract

Comparing language-model variants requires separating weight-associated differences from variation introduced by the probe data. We introduce nDNA, a task-calibrated metric that compares layerwise next-token prediction profiles using Fisher–Rao length, a supervised correction-field norm and constrained dynamic time warping. Across four decoders and ten tasks, length is strongly decoder-associated (ω^2=0.83; 40/40 same-decoder nearest neighbours), whereas the field is primarily task-associated (ω^2=0.64). When each of three task families is held out in turn, 35/40 profiles identify the correct decoder and 37/40 remain within its fitted task reference. Length identifies 18/18 models across independently sampled 1,000- and 2,500-text SQuAD sets. In a metadata-controlled cohort, length and field retrieve 15/18 related models (AUC 0.86; MRR 0.88), compared with metadata AUC 0.50. Under frozen task-only references, all four 4-bit variants remain below threshold and all four 40% pruning variants exceed it. By generation 8, recursive self-training reduces distinct-3 from 0.711 to 0.501 and raises held-out perplexity from 7.12 to 8.23, versus 0.684 and 7.34 under a human-data control; aligned cost from generation 0 is 0.108 versus 0.039. Spectral curvature is a secondary descriptor. nDNA is observational, white-box and not a model-quality or causal measure.

1 Introduction

Fine-tuning, alignment, quantisation, pruning, merging and distillation modify language models in different ways. Their effects are usually evaluated through endpoint behaviour, such as perplexity, task accuracy or refusal rates. These measurements are essential, but they do not describe how intermediate predictions evolve through the network. Two checkpoints with similar endpoint scores may therefore still differ in the depthwise organisation of their predictive computation.

Intermediate next-token readouts provide a way to examine this organisation. For a fixed input position, applying the output head at successive layers (the logit lens) produces an ordered sequence of predictive distributions. Rather than treating each layer independently, we study this sequence as a path in probability space. The resulting profile describes where predictions change substantially, where successive changes alter direction, and how a supervised target induces a local correction signal. The path depends on the chosen readout, the temperature and the vocabulary, which we keep fixed within a comparison.

A central difficulty is that the profile also depends on the examples used to measure it. A difference between two checkpoints can therefore be interpreted only relative to the variation observed when the probe task changes within one checkpoint. We ask: *can layerwise prediction profiles give a useful comparison of language-model checkpoints once the variation introduced by the probe tasks is accounted for?*

Approach and scope. nDNA is a model-comparison metric built from a profile representation and a comparison rule. The representation summarises layerwise predictive movement by the *thermodynamic length* (the Fisher-Rao path length) and the correction induced by a supervised target by the *belief vector field* (a supervised correction field); *spectral curvature* measures the turning of the path and is treated as a secondary geometric descriptor (App. B). The protocol specifies how profiles are scaled and aligned with constrained dynamic time warping (DTW), and when a checkpoint difference can be compared with a *task reference*, the spread of distances between the same model's profiles on different tasks (Fig. 1). The geometric definitions establish what is measured; the empirical protocol determines which comparisons are meaningful. Behaviour is measured separately

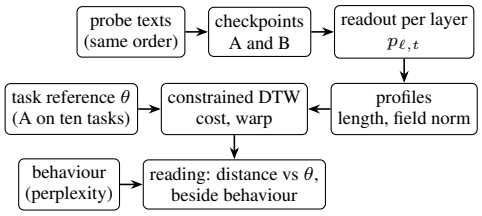


Figure 1: **The protocol.** Measurement (readout and profiles), comparison (DTW) and decision (task reference) are separate steps; behaviour is measured beside them. A low profile distance does not establish equal behaviour.

because profile similarity and behavioural equivalence are different questions. The framework is an observational, white-box diagnostic whose task reference is specific to the evaluated probes, scalings and models.

Findings and contributions. Across four decoders and ten tasks, length profiles are associated with the decoder and field-norm profiles with the task, so the channels need to be reported separately and interpreted against task variation. This pattern persists when the early layers are removed, when both channels use the same positions, and in a tuned-lens evaluation. Profiles identify checkpoints across independent samples and retain decoder information when an entire task family is held out. The operation studies then show where the measurement is informative and where it is not: compression severity changes both profiles and behaviour, recursive self-training contracts the path as diversity falls, and the evaluated alignment stages are not consistently separated. The contributions are (i) formal definitions and analytical properties for three prediction-path descriptors, with length as the primary empirical channel, the field as a complementary conditioned channel, and curvature as a secondary descriptor (Sec. 3); (ii) a protocol that separates measurement, comparison and decision (Sec. 4); and (iii) a model-comparison study covering task calibration, identification and bounded operation case studies (Tab. 1).

2 Related Work: Prediction Readouts and Model Comparison

Layerwise prediction readouts. Layerwise prediction readouts provide the starting point for our analysis. The logit lens applies the output head to intermediate layers (nostalgebraist, 2020); the tuned lens learns translators that correct its early-layer bias (Belrose et al., 2023); and related readouts shortcut layers (Din et al., 2023), forecast later tokens (Pal et al., 2023) or decode states in natural language (Ghandeharioun et al., 2024). Intermediate predictions have also been used to identify a critical layer under false demonstrations, to trace latent-language transitions and to interpret feed-forward updates in vocabulary space (Halawi et al., 2023; Wendler et al., 2024; Geva et al., 2022). Other work describes depth as a sequence of prediction phases (Lad et al., 2024; Lombardo et al., 2026), while layer-removal and reordering studies show that depth roles need not be unique or uniformly necessary (Gromov et al., 2024; Sun et al., 2024; Skean et al., 2025). These methods expose what a layer predicts; we study the ordered sequence of those readouts. The closest work is Jung and Jung (2026), who treat adjacent-layer logit updates as vectors and summarise their coherence and angular rotation over a dataset to explain model behaviour. We instead measure the change between successive predicted distributions with the Fisher-Rao metric and compare the resulting profiles across checkpoints against a task reference. We do not run their logit-update representation on our cohorts, so this is a conceptual rather than empirical comparison. Applying the final output head to early layers can also misrepresent information encoded in those states.

Model similarity and lineage. Several families of methods compare models, and each answers a different question. Representation similarity measures such as CKA and SVCCA compare activation spaces (Kornblith et al., 2019; Raghu et al., 2017; Klabunde et al., 2025), and their values change with the data used to measure them (Ciernik et al., 2024), which is the problem our task reference addresses for prediction profiles. Provenance methods infer lineage from outputs (Yax et al., 2025), weights (Zeng et al., 2024), layerwise representations (Zhang et al., 2025) or initialisation signatures (Tong et al., 2026). Patching, causal scrubbing, circuit discovery and sparse autoencoders ask which component or feature produces a behaviour (Heimersheim and Nanda, 2024; Goldowsky-Dill et al., 2023; Chan et al., 2022; Conmy et al., 2023; Cunningham et al., 2023), a question our profiles do not address. Prior studies have found that some forms of fine-tuning and alignment modify existing mechanisms or concentrate their changes in a limited set of behaviours (Prakash et al., 2024; Jain et al., 2024; Lee et al., 2024; Qi et al., 2025).

Prediction-space geometry. The Fisher metric is the invariant metric on statistical models (Chentsov, 1982; Amari, 2016; Ay et al., 2017; Nielsen, 2020), with a closed-form distance between categorical laws (Rao, 1945; Bhattacharyya, 1943). Thermodynamic length is path length in this metric (Weinhold, 1975; Salamon and Berry, 1983; Crooks, 2007), and turning angle has been used to measure the straightness of perceptual and sentence trajectories (Hénaff et al., 2019; Hosseini and Fedorenko, 2023). The closest scalar baseline is the divergence between adjacent layers, with which thermodynamic length agrees locally (Thm. 1(iii)) while being symmetric and additive along depth (App. S). Our aim is therefore complementary to representation similarity, causal intervention and provenance fingerprints: we ask whether a depth-resolved prediction profile with an explicit task reference usefully describes checkpoint differences, on curated cohorts and without a head-to-head comparison against provenance methods.

3 Layerwise Prediction Profiles

Readout and sampling. For a teacher-forced prefix $x_{\leq t}$ and a selected transformer-block output indexed by ℓ (the embedding output is excluded), we apply the model's final normalisation and output head to obtain logits $z_{\ell,t} \in \mathbb{R}^V$ and

$$p_{\ell,t} = \mathrm{softmax}(z_{\ell,t}/\tau) \in \mathring{\Delta}^{V-1}, \quad u_{\ell,t} = \sqrt{p_{\ell,t}}, \tag{1}$$

with $\tau{=}1$ throughout. The target at position t is the next token $y_t^\star$. For the finite profiles reported here, $\mathbb{E}$ denotes the arithmetic mean over retained positions. A prompt with more retained positions therefore receives more weight; which positions are retained differs between experimental blocks and is listed in Tab. 1. The map $p \mapsto 2\sqrt{p}$ is an isometry from the simplex with the Fisher metric onto the positive orthant of the sphere of radius two (Prop. 3), so

$$d_{\mathrm{FR}}(p,q) = 2\arccos\langle\sqrt{p},\sqrt{q}\rangle \in [0,\pi]. \tag{2}$$

Profiles of models with different tokenisers are computed on different supports, and we do not claim they share a scale. Proofs are in Apps. C to F.

3.1 Thermodynamic length (Fisher-Rao path length)

The length profile records the magnitude of predictive change between successive readouts. With $d_{\ell,t} = d_{\mathrm{FR}}(p_{\ell,t}, p_{\ell+1,t})$, the length of one path and the layer profile are

$$\mathrm{TL}(x,t) = \sum_{\ell=0}^{L-1} d_{\ell,t}, \qquad \mathcal{L}_\ell = \mathbb{E}\,[d_{\ell,t}]. \tag{3}$$

Theorem 1 (Length). *For a fixed position write $p_\ell = p_{\ell,t}$ and $d_\ell = d_{\ell,t}$.*

(i) $d_{\mathrm{FR}}(p_0, p_L) \leq \mathrm{TL}$, *with equality iff $p_0, \ldots, p_L$ lie in this order on the minimising geodesic from p_0 to p_L.*

(ii) For every ℓ, $d_{\mathrm{FR}}(p_\ell, p_L) \leq \sum_{k=\ell}^{L-1} d_k$, hence $\mathbb{E}\, d_{\mathrm{FR}}(p_{\ell,t}, p_{L,t}) \leq \sum_{k\geq\ell} \mathcal{L}_k$. The converse fails.

(iii) $\mathrm{TL}^2 \leq L\sum_k d_k^2$, *with equality iff all d_k are equal; and, for fixed L, if the relative steps $\varepsilon_k = \max_i |p_{k+1,i} - p_{k,i}|/p_{k,i}$ satisfy $\max_k \varepsilon_k \to 0$,*

$$\textstyle\sum_k d_k^2 = 2\sum_k D_{\mathrm{KL}}(p_k \| p_{k+1})\,\big(1 + o(1)\big).$$

Summing the steps gives the length of the piecewise-geodesic path through the layerwise distributions, which distinguishes a direct transition from a sequence of revisions that reaches the same endpoint. It does not distinguish useful revisions from changes that are later reversed: part (ii) bounds how far the prediction can still move after layer ℓ, but it is a distance bound and not a measure of influence. The relative-step condition in (iii) matters for vocabularies with many tiny probabilities.

3.2 Belief vector field (supervised correction field)

The belief vector field introduces supervision into the profile. At each layer we compute the change in the square-root distribution induced by a logit-gradient step toward the supplied target token. The logit gradient of the target's log-probability, pushed through $z \mapsto \sqrt{\mathrm{softmax}(z/\tau)}$, is

$$g_{\ell,t} = \nabla_z \log p_{\ell,t}(y_t^\star) = \tau^{-1}\big(e_{y_t^\star} - p_{\ell,t}\big), \tag{4}$$

$$w_{\ell,t} = \tfrac{1}{2\tau}\big(I - u_{\ell,t}u_{\ell,t}^\top\big)\,\mathrm{Diag}(u_{\ell,t})\,g_{\ell,t}. \tag{5}$$

The *belief push* $w_{\ell,t}$ is the image of a Euclidean gradient step in logit space; it depends on the labels and on τ. Because pushes are based at different points on the sphere, we transport them to $\bar{u}_\ell = \mathbb{E}[u_{\ell,t}]/\|\mathbb{E}[u_{\ell,t}]\|$ by parallel transport, $\mathrm{PT}_{u\to a}(v) = v - \frac{\langle v,a\rangle}{1+\langle u,a\rangle}(u+a)$, and average over a condition c:

$$\mathbf{v}_\ell(c) = \mathbb{E}_{(x,t)\sim P(c)}\big[\,\mathrm{PT}_{u_{\ell,t}\to\bar{u}_\ell}(w_{\ell,t})\big]. \tag{6}$$

Theorem 2 (Field). *Write $p = p_{\ell,t}$, $u = u_{\ell,t}$, $w = w_{\ell,t}$ and $y = y_t^\star$.*

(i) $\langle w, u\rangle = 0$, w is unchanged by $z \mapsto z + c\mathbf{1}$, and

$$4\tau^4\|w\|^2 = p_y(1-2p_y) + \textstyle\sum_i p_i^3 - \left(p_y - \sum_i p_i^2\right)^2,$$

which is positive for interior p when the vocabulary has at least two tokens, at most 1, and tends to 0 as p approaches any vertex, correct or not.

(ii) If $1 + \langle u_{\ell,t}, \bar{u}_\ell\rangle > 0$ for every retained position, parallel transport is well defined and isometric. Consequently, $\|\mathbf{v}_\ell(c)\| \le \mathbb{E}\|w_{\ell,t}\|$, with equality iff all transported pushes are non-negative multiples of one vector.

The norm of the average reflects both the strength of the individual corrections and their directional agreement, so a small field norm can mean weak pushes, confident predictions (right or wrong) or pushes that cancel. The retained runs use the projected ambient mean $\widetilde{\mathbf{v}}_\ell = \mathrm{Proj}_{\bar{u}_\ell}\mathbb{E}[w_{\ell,t}]$ rather than the transported mean in Eq. (6). Their difference is bounded by $\frac{1}{2}\mathbb{E}[\|w\|\|u - \bar{u}_\ell\|^2]$ (Prop. 7), but the spread term was not retained and the numerical discrepancy is therefore unknown. Empirical comparisons use $\|\widetilde{\mathbf{v}}_\ell\|$ and discard direction; $\mathbf{v}_\ell$ denotes the exact transported construction.

3.3 Spectral curvature as a secondary descriptor

A third descriptor, spectral curvature, records whether successive predictive changes continue in one direction or reverse: the exterior turn between adjacent geodesic segments divided by their mean Fisher-Rao length. We define it, prove that it depends only on Fisher-Rao distances, and show that it converges to Fisher-Rao geodesic curvature (Thm. 4, App. F), and check the estimator on simulated paths of known curvature, where it is accurate to 0.2% with 16 layers. A matched model-scale evaluation gives corrected curvature $\omega^2 = 0.31$ for decoder and 0.21 for task, 29/40 same-decoder nearest neighbours, 13/18 cross-sample identifications and retrieval AUC 0.75 (Tab. 26). Fisher-Rao length gives 0.95, -0.15, 40/40, 18/18 and 0.91 on the same quantities, so curvature is retained as a secondary channel rather than the default comparator. Earlier three-channel exports used an interior-angle score, which behaves mainly as inverse step size (correlation 0.80 to 0.90 with $\pi/(a+b)$); results from those exports remain labelled separately.

4 Comparison and Decision

Comparison by constrained DTW. A peak at layer 22 in one profile and at layer 24 in another may be the same event, which a layer-by-layer comparison scores as disagreement. Let $\mathbf{z}_i, \mathbf{z}'_j$ stack the channels used at depth points i and j after a stated scaling, and let Π_w be the monotone, continuous alignment paths from $(1,1)$ to (L,L') inside the band $|i/L - j/L'| \le w$ (Sakoe and Chiba, 1978; Müller, 2007), $w = 0.2$. We minimise the accumulated cost and then divide by the length of the chosen path:

$$\pi^\star = \arg\min_{\pi\in\Pi_w} \textstyle\sum_{(i,j)\in\pi}\|\mathbf{z}_i - \mathbf{z}'_j\|^2,$$

$$D_{\mathrm{RMS}} = \left(\frac{1}{|\pi^\star|}\sum_{(i,j)\in\pi^\star}\|\mathbf{z}_i - \mathbf{z}'_j\|^2\right)^{1/2}. \quad (7)$$

which is not the same as minimising the mean cost. Channels enter as separate coordinates, so the cost adds their squared differences; we do not multiply, add or subtract them into one number, because no continuous per-layer combination of two channels is injective and every such combination loses identification and retrieval on our profiles (App. T). The *warp* $\mathrm{W} = |\pi^\star|^{-1}\sum|i-j|$ is measured in depth points: native layers for native-depth profiles and thirty-seconds of depth for retained profiles resampled to 32 points. For unequal depths we instead report $|\pi^\star|^{-1}\sum_{(i,j)\in\pi^\star}|(i-1)/(L-1) - (j-1)/(L'-1)|$. DTW is a dissimilarity rather than a metric; we interpret values through matched comparisons rather than metric-space guarantees.

Decision by a task reference. For a model profiled on ten tasks, the *task reference* is the set of DTW distances between its task profiles, and the *threshold* is its 0.9 quantile over the $\binom{10}{2} = 45$ unique task pairs, $\theta = Q_{0.9}\{\mathrm{DTW}(\bar{\Gamma}_s, \bar{\Gamma}_r) : s < r\}$. A comparison is read as "above task variation" only if it uses the same probe, positions and scaling as θ; $Q_{0.9}$ uses linear interpolation between adjacent ordered values. The 45 pairs share ten tasks and are not independent, and there is one seed, so θ is a descriptive, model-specific reference for this probe suite and not a significance test.

What is invariant. The descriptors depend on hidden states only through the predicted distributions and not on how the simplex is parametrised

(Prop. 2), but they do depend on the readout, τ, the vocabulary and the scaling. This invariance is analytic and was not confirmed empirically.

5 Experimental Design and Evidence Scope

We ask three questions. **RQ1**: how do task changes affect each channel? **RQ2**: do profiles retain checkpoint-associated information across a change of sample size and across related checkpoints? **RQ3**: what profile differences appear for selected operations, and how do they relate to behaviour? Tab. 1 identifies the probe, sampling unit, estimator and scaling for each answer. Per-layer arrays support RMS-DTW reanalysis for the task, identification and retrieval studies. Aggregate-only operation results retain their reported aligned mean-squared cost and are not compared numerically with RMS-DTW. The decomposition cohort (Gemma-3 1B PT and IT, Llama-3 8B and Qwen-3 4B) differs from the four-decoder threshold cohort (Gemma-3 1B, Llama-3.1 8B and Qwen-3 4B and 8B). Empirical field results use $\|\widetilde{\mathbf{v}}\|$, the projected estimator, and do not validate field direction or exact parallel transport.

Claim vocabulary. "Identified" means that a query profile's nearest member of a fixed candidate set is the same checkpoint; "related" means that two checkpoints belong to a documented ancestry group, which may span related base releases and does not imply that one is the immediate parent. "Task-calibrated" applies only when the comparison uses the probe, positions, channels and scaling that define θ. We call an array-derived quantity measured and reserve interpretation for what that quantity may imply; neither word denotes causal evidence.

6 Task Variation and Channel Sensitivity

The descriptors define the profile but do not determine how much of a difference is attributable to the measurement conditions, so we begin with task sensitivity. A change of probe task can alter a checkpoint's profile even when its weights are fixed; we measure this variation within each decoder and use the distances between its task-specific profiles as an empirical reference for the evaluated suite.

6.1 Decoder and Task Associations Across Channels

We first ask whether each channel is more strongly associated with the decoder or with the task. For 40 profiles (four decoders, each on the ten tasks) we fit a decoder factor and, separately, a task factor to the DTW distance matrix of each channel set with a permutational analysis of variance (Anderson, 2001), and report the bias-adjusted $\omega^2 = (\mathrm{SS}_B - (k-1)\mathrm{MS}_W)/(\mathrm{SS}_T + \mathrm{MS}_W)$ (Tab. 2 and Fig. 2). The two fits are separate and their values do not add. Because every decoder is observed on every task, the primary permutation test shuffles decoder labels within tasks and task labels within decoders. This preserves the crossed design while testing one factor conditional on the other.

In the four-decoder, ten-task cohort, length profiles show strong decoder-associated structure: under RMS-DTW the decoder gives $\omega^2 = 0.83$ and every profile's nearest neighbour belongs to the same decoder (40 of 40). The adjusted task estimate is negative (-0.19), which is compatible with no detectable one-way task effect rather than negative variance. A restricted shuffle nevertheless detects a smaller task association ($p = 0.002$), and on the same task two decoders lie a median 3.7 times farther apart than two tasks within one decoder (Tab. 22). Field-norm profiles show the opposite pattern: $\omega^2 = 0.64$ for the task against 0.15 for the decoder, as expected of a quantity computed from the supervised targets. Length and field together give 0.32 for the decoder and 0.37 for the task, and for a close pair (Llama-3 8B and R1-Distill-Llama 8B over four datasets) decoder and dataset effects are of the same order (App. J); without rescaling the pair separates by model on all three datasets, so per-profile scaling removes the largest sibling difference (App. L). The decoder association does not disappear when the early layers, where the logit lens is least reliable, are removed: on the last two thirds of depth length remains strongly decoder-associated (App. K). It also survives two readout checks. Matching length and field to the same last eight positions changes their decoder/task associations only from 0.94/$-$0.10 to 0.93/$-$0.06 for length and from 0.17/0.85 to 0.18/0.88 for the field. On a 30-profile, 12-checkpoint subset, a tuned lens gives decoder $\omega^2 = 0.88$, 29/30 same-decoder neighbours and 12/12 cross-sample identifications, against 0.91, 30/30 and 12/12 for the logit lens (Tabs. 27 and 28). Field coherence rises from 0.33

Table 1: **Experiments and comparison contracts.** A threshold is applicable only inside the block that fitted its task-only scaler. Length uses all loss-bearing positions in the retained dashboard profiles and the field uses the last eight; a matched-position analysis is reported in Tab. 28. Corrected curvature is evaluated separately as a secondary descriptor and is not used in the primary thresholds.

Experiment	Probe and sample	Channels	Scaling	θ
Channel sensitivity (RQ1)	10 tasks × 4 decoders	$\mathcal{L}$; $\|\tilde{\mathbf{v}}\|$	task profiles only	defines
Corrected-curvature validation	40 known-curvature paths; matched 40-profile cohort	κ	matched evaluation	no
Held-out task families (RQ1)	3 fixed folds, 40 held-out profiles	$\mathcal{L}$, $\|\tilde{\mathbf{v}}\|$	training families only	tests
Independent-sample identity (RQ2)	SQuAD, 1000 vs 2500 random texts, 18 models	$\mathcal{L}$	per profile	no
Controlled related retrieval (RQ2)	18 queries; metadata-balanced candidates	$\mathcal{L}$, $\|\tilde{\mathbf{v}}\|$	per profile	no
Compression severity (RQ3)	4 decoders; quantisation and pruning	$\mathcal{L}$, $\|\tilde{\mathbf{v}}\|$	frozen task-only	yes
SFT/DPO alignment resolution (RQ3)	Litmus and HarmBench; 2 families	$\mathcal{L}$, $\|\tilde{\mathbf{v}}\|$	within probe	no
Distillation profile comparison (RQ3)	2 role-verified teacher–student pairs	$\mathcal{L}$, $\|\tilde{\mathbf{v}}\|$	within pair	no
Recursive self-training (RQ3)	128 prompts; 512 evaluation texts	$\mathcal{L}$ and behaviour	none	no

Table 2: **Association with decoder and with task, per channel set** (40 profiles; separate fits; restricted permutation p from 2000 shuffles; last two columns: the nearest profile shares the model or the task). Profiles are rescaled jointly over each model's ten tasks, so task differences keep their amplitude and decoder differences concern shape. Free-permutation and original-angle results appear in Tab. 22.

	decoder		task		nearest profile		
Channels	ω^2	p	ω^2	p	model	task	ratio
$\mathcal{L}$ only	0.83	<0.001	-0.19	0.002	40/40	0/40	3.7×
$\|\tilde{\mathbf{v}}\|$ only	0.15	<0.001	0.64	<0.001	31/40	4/40	0.7×
$\mathcal{L}$, $\|\tilde{\mathbf{v}}\|$	0.32	<0.001	0.37	<0.001	39/40	1/40	1.1×

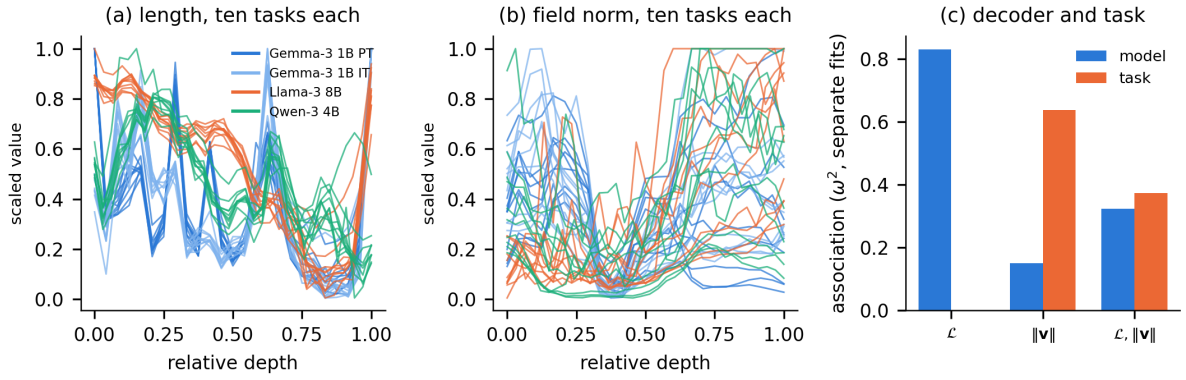


Figure 2: **Decoder and task associations by channel.** (a, b) Length and field-norm profiles of four decoders on ten tasks each (colour is the decoder). Length profiles show stronger separation by decoder than by task in this cohort, and field profiles stronger separation by task. (c) Bias-adjusted association for the three curvature-free channel sets; decoder and task effects are fitted separately and are non-additive.

in the first depth third to 0.60 in the last, so the aggregate field increasingly reflects aligned rather than cancelling token-level pushes (Tab. 29). These results motivate reporting the channels separately and calibrating any combined distance against task variation.

This calibration also transfers imperfectly to tasks excluded from fitting. We group the ten tasks by evaluation format and hold out each of three families in turn. The channel scaler, decoder references and task thresholds are fitted on the remaining families only. Across the resulting 40 held-out profiles, the nearest calibration set belongs to the correct decoder for 35, and 37 lie within their decoder's

fitted task reference (Tab. 18). The generation fold is hardest (13/16 on both criteria). This is a fixed leave-family-out check on the retained tasks, not evidence for arbitrary unseen domains.

6.2 A Task Reference for Each Model

Using RMS-DTW on length and field, task-only thresholds in the decomposition cohort range from 0.384 to 0.437 (Tab. 19). Their variation across models is why we do not pool them into a universal cutoff. The fixed-family analysis above is the more demanding check: three held-out profiles exceed their decoder's calibration reference, although decoder identity remains correct for two of those three. The threshold is therefore an empirical reference for a specified decoder, probe family, position rule and scaler, not a significance boundary.

The operation cohort uses a separately retained task-only scaler that was frozen before variants were evaluated. In that block, 4-bit distances are 0.002–0.011 against thresholds 0.014–0.050, 40% pruning is 0.147–0.231, and the largest sibling distance is 0.003–0.024 (Tab. 30). These values are reported aligned mean-squared costs, not RMS-DTW, and are compared only within that block. Fisher–Rao length gives the intended quantisation, pruning and sibling decision in all 13 comparisons, against 10 of 13 for forward KL (Tab. 26).

7 Checkpoint Identification and Related-Checkpoint Retrieval

Having characterised task-associated variation, we ask whether profiles retain enough checkpoint-associated structure to support identification and related-checkpoint retrieval.

The same checkpoint across sample sizes. Eighteen open models were profiled on separately sampled random SQuAD sets of 1000 and 2500 texts; the smaller set is not a prefix of the larger. With length alone, the nearest among the same 18 candidates is the same checkpoint for all 18 (median margin to the nearest other model 2.8, minimum 1.38). Adding the field lowers this to 14 of 18, as expected of the more sample-sensitive channel (Tab. 36). Architecture metadata (depth, width, vocabulary, tokenizer) would identify 12 of 18 in expectation, because siblings share a configuration, and on the last two thirds of depth length still identifies 18 of 18 (App. K). This tests identification, not whether θ changes with n, and per-model scaling removes amplitude.

Table 3: **Related-checkpoint retrieval** (length and field). Nearest neighbour is a relative (models with a relative only), chance rate, permutation p (3000 shuffles), AUC (related pairs closer than unrelated) and mean reciprocal rank. Other channel sets: Tab. 46.

Cohort	N	NN	chance	p	AUC	MRR
n=2500	19	14/14	0.13	<0.001	0.95	1.00
n=1000	18	13/13	0.11	<0.001	0.93	1.00
19-model set	19	18/19	0.11	<0.001	0.93	0.96

Related checkpoints. We group models by documented ancestry (Llama-2, Llama-3, Qwen1.5, DeepSeek-LLM and Gemma families; five models have no relative; App. O), so groups are families of related checkpoints and can span related base releases. They are not parent-child pairs. For each model with a relative, we ask whether its nearest neighbour among all other models in the cohort is a relative (leave one out). Models without a relative are not in the denominator.

At n=2500 the nearest neighbour is related for 14 of 14 eligible models (chance 13%, $p < 0.001$, AUC 0.95), versus 10 of 14 for length alone; at n=1000 the result is 13 of 13. A second 19-model cohort gives 18 of 19 (Tab. 3). These curated results are partly explained by architecture metadata (Tab. 25) and do not establish exact parentage or open-set provenance. We therefore also evaluate a candidate set in which the selected metadata are uninformative by construction. Metadata gives AUC 0.50, whereas length and field retrieve 14/18 queries at n=1000 and 15/18 at n=2500 (AUC 0.86, MRR 0.88; Tab. 31). The controlled cohort remains fixed and non-adversarial; broader maps, depth ablations and seven distance functions are reported in Apps. G and O.

8 Operation Case Studies: Profile Change and Behaviour

The operation studies cover quantisation, pruning, SFT/DPO alignment, distillation and recursive self-training. They ask which transformations produce visible profile differences and whether those differences accompany behavioural degradation. Only quantisation, pruning and siblings use θ; the other cases are descriptive (Tab. 39 and App. P).

Compression and siblings. With a frozen task-only scaler, all four 4-bit variants and five published siblings remain below θ, while all four 40% pruning variants exceed it (Tab. 30). In a separate

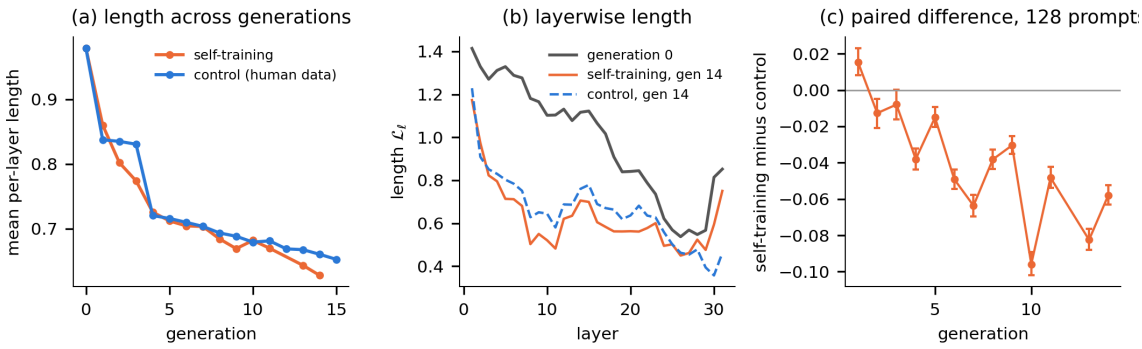


Figure 3: **Recursive self-training and the length profile.** (a) Mean length across self-training and human-answer control generations. (b) The layerwise profiles contract in both runs. (c) On 128 unpadded prompts, self-training has shorter steps than the control from generation 2 onward; bars are 95% prompt-bootstrap intervals. A padding change prevents interpreting panel (a) as a controlled difference.

compression cohort, median distance rises from 0.014 for 4-bit quantisation to 0.061 for 20–30% and 0.113 for 40–50% pruning; mean benchmark loss rises from 0.9 to 4.1 and 9.6 points (Tab. 32). This is a severity association in a small block, not a universal compression test.

Alignment stages. On Litmus and HarmBench, model family explains the pooled distance structure ($R^2 = 0.98$), whereas SFT/DPO stage does not ($R^2 = 0.002$). Two additional SQuAD profiles concentrate their differences late in depth, but this is a resolution-limit result, not evidence that alignment has no effect (Sec. P.3).

Distillation. In two role-verified math-distillation pairs, the trained student remains closer to its initial checkpoint than to its teacher: aligned costs are 0.107 versus 0.164 for Llama and 0.190 versus 0.304 for Qwen (App. P). Without a matched behavioral test, this is a profile-similarity result rather than evidence for or against capability transfer.

Recursive self-training. Recursive self-training uses generated answers as the next generation's fine-tuning data; "collapse" here means the accompanying degradation in diversity and held-out perplexity, not contraction alone. Across generations 0–8, self-training changes distinct-3 from 0.711 to 0.501, self-BLEU from 0.326 to 0.512 and WikiText-103 perplexity from 7.12 to 8.23; the human-data control ends at 0.684, 0.359 and 7.34. Length ends at 0.631 versus 0.702, and distance from generation 0 at 0.108 versus 0.039 (Fig. 3 and Tab. 49). Nine additional trajectories spanning 10–17 checkpoints contract by 17%–83%, but lack matched behavioural records (App. Q).

9 Discussion and Conclusion

The three channels answer different questions. Thermodynamic length is the default comparison when the aim is checkpoint identity or decoder-associated profile shape. The belief-vector-field norm is useful when the target-conditioned response is itself of interest, but it also increases task sensitivity. Spectral curvature describes turning and is mathematically distinct, yet it is weaker than length on the matched checkpoint tests. A high nDNA distance establishes a change in the selected layerwise readout under a specified probe and scaler. A low distance does not establish equal quality, behavior or mechanism.

Within those limits, the study supports a narrow conclusion: task-calibrated prediction profiles provide a useful white-box diagnostic for comparing language-model variants. Length identifies models across independent samples, held-out task families retain substantial decoder information, and controlled retrieval remains above its metadata baseline. Compression and recursive self-training show how profile changes can be read beside behavioral outcomes. Larger controlled cohorts, exact-field evaluation, multiple seeds and causal tests are the principal next validations.

10 Limitations

Measurement validity. The logit lens can be unreliable in early layers. Here, the decoder effect survives early-layer exclusion and a 30-profile tuned-lens replication, but broader cross-readout validation is still needed. Corrected curvature is

evaluated on known-curvature paths and matched model profiles; the exact transported field remains unevaluated. The dashboard's all-position length and last-eight field are also checked with both restricted to the last eight positions. **Comparability.** Profiles depend on the readout, temperature, tokeniser and scaling. Per-profile dashboard scaling retains shape only, while groupwise scaling also retains relative amplitude within that group; blocks scaled differently are not compared with each other. **Inference.** Small cohorts (four decoders, ten tasks, two retrieval cohorts), one seed and no prompt-resampling intervals outside recursive self-training; the 45 task pairs are not independent. The leave-family-out check uses a fixed partition of the same ten retained tasks and does not cover arbitrary new domains. **Scope of claims.** Observational and pooled over positions; no feature identity or causal validation, and the field's direction is not evaluated. Compression and recursive training include behavioural outcomes, but the other operation cases remain profile-only and no release decision rule is validated. **Access.** White-box, decoder-only models (App. U).

Reproducibility statement

`analysis.py` regenerates the derived tables and figures from the retained artifacts: JSON exports of four decoders (`results/`) and per-layer profiles decoded from the experiment dashboards (`results/dashboard_profiles.json`), the recursive runs exported with a SHA-256 hash of every source file (`export_recursive_runs.py`, `results/recursive_runs.json`), the unscaled sibling-pair profiles (`export_closepair_raw.py`, `results/closepair_raw.json`) and the SFT and DPO length profiles of two public dashboards (`results/alignment_sft_dpo.json`). `extract_collapse_html.py` independently decodes the nine additional recursive trajectories, records source hashes and regenerates their JSON summaries and figure. The matched baseline, readout, frozen-scaler, coherence, compression and recursive-behaviour aggregates are recorded in `results/additional_matched_results.json`. The experiment drivers and both estimator versions (original interior angle and corrected exterior turn) are in the accompanying repository. The four-decoder JSON export manifests record model identifiers, seeds, probe names, prompt counts, timestamps and code commits; absolute-profile exports also retain probe hashes. The recorded environment pins PyTorch 2.9.0 and Transformers 4.57.3. Runtime hardware, peak memory, Hub revision hashes for every checkpoint and the sample sizes of the ten-task dashboards are not present in the retained result manifests. This supports reproduction of the reported analysis from retained profiles and aggregates. It does not by itself reproduce model inference for blocks whose per-position outputs were not retained.

Ethics statement

This work measures the internal computation of public language models and introduces no new capability. Checkpoint-similarity measurements may inform provenance investigations but are not evidence of ownership or of undisclosed training. The culture-specific fine-tuned variants are measurement instruments, not deployable systems, and a regional label names a training corpus, not a culture.

AI use statement

We used generative AI tools for writing and coding assistance. The authors verified the derivations, the code and the retained evidence, and take responsibility for the content.

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Table 4: **The ten stability tasks.**

Task	Type	Source
AI2-ARC	science question answering	Clark et al. (2018)
HellaSwag	commonsense completion	Zellers et al. (2019)
WinoGrande	coreference	Sakaguchi et al. (2020)
MNLI	natural language inference	Williams et al. (2018)
QQP	paraphrase	Wang et al. (2019)
SQuAD v2	extractive QA with unanswerables	Rajpurkar et al. (2018)
CNN/DailyMail	summarisation	See et al. (2017)
CommonGen	constrained generation	Lin et al. (2020)
WMT16	translation	Bojar et al. (2016)
IMDB	sentiment	Maas et al. (2011)

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A Experimental setup and artifacts

Probe sets. Task stability uses ten FLAN-style tasks (Wei et al., 2022) (Tab. 4), each with its own prompt distribution. Dashboard runs use SQuAD (Rajpurkar et al., 2016) at 1000 or 2500 texts per model, truncated at 128 tokens. The alignment ladder uses the Litmus safe and unsafe prompts and HarmBench (Mazeika et al., 2024); the distillation runs use GSM8K (Cobbe et al., 2021) and SQuAD; the two-family sibling test uses AG News (Zhang et al., 2015), AutoMathText (Zhang et al., 2024b), SQuAD and the Stanford Encyclopedia of Philosophy. For every comparison the base and the modified model are profiled on identical prompts in identical order. The exports record the model identifier, seed, number of prompts, estimator setting and code commit; the absolute-profile exports also contain probe hashes. Hub revision hashes for each checkpoint are not present in the retained result manifests.

Models and operations. The four main decoders are Gemma-3 1B (Gemma Team, 2025), Llama-3.1 8B (Llama Team, AI @ Meta, 2024) and Qwen-3 4B and 8B (Yang et al., 2025). Quantisation is round-to-nearest 4-bit (and 2-bit in the failure controls); pruning is structured magnitude pruning at 30, 40 and 60%; siblings are the published instruct, reasoning or paired checkpoints. The dashboard suite contains Llama-2 (Touvron et al., 2023), Llama-3 and Tulu 3 (Llama Team, AI @ Meta, 2024; Lambert et al., 2024), Vicuna (Chiang et al., 2023), Orca 2 (Mitra et al., 2023), Mistral (Jiang et al., 2023), Qwen (Bai et al., 2023),

DeepSeek (DeepSeek-AI, 2024, 2025), Gemma (Gemma Team, 2024), Falcon (Almazrouei et al., 2023), Phi-2 (Javaheripi and Bubeck, 2023), Pythia (Biderman et al., 2023), TinyLlama (Zhang et al., 2024a) and Olmo (Groeneveld et al., 2024). Regional adapters are LoRA (Hu et al., 2022) fine-tunes of one Qwen-3 base and their 28 pairwise merges use Fisher-weighted averaging (Matena and Raffel, 2022). Alignment checkpoints are SFT and DPO (Rafailov et al., 2023) versions trained with the project scripts. Distillation is LoRA training with a KL term to a teacher (Hinton et al., 2015); Low-Rank Clone students (Hao et al., 2025) appear in the distillation dashboards.

Artifacts. `results/` holds the 32 JSON exports of four archived experiments (behaviour versus geometry, task stability, patching, absolute profiles; four models, one seed) and the decoded dashboard profiles (`dashboard_profiles.json`); `analysis.py` regenerates `results/derived.json`, `tables.tex` and all `figures/fig_*.png`. Dashboard extraction parses each embedded Plotly `newPlot` payload, decodes binary `bdata` with its declared dtype (including float32 and float64), and copies each channel's plotted y array into JSON. No rounding is applied during extraction; rounding occurs only in displayed tables. The retained precision is therefore the precision stored in the dashboard, but the original per-prompt values cannot be recovered from these aggregate arrays. The experiment drivers (`core/`, `experiments/`) and their model-free tests are in the accompanying repository.

Dashboard scaling. The SQuAD n=1000 and n=2500 profiles are min-max scaled separately to $[0.05, 1]$. The second SQuAD cohort, merges and four-dataset sibling comparison are clipped at $Q_1 - 1.5\,\mathrm{IQR}$ and $Q_3 + 1.5\,\mathrm{IQR}$ within each channel and profile, then scaled to $[0.005, 1]$. The ten-task, alignment, distillation and HarmBench profiles use the same clipping rule fitted jointly within each comparison group. Per-profile scaling retains clipped relative shape only; groupwise scaling also retains relative amplitude within that group, but not across groups. Clipping can change local shape as well as amplitude.

B Terminology

nDNA is the model-comparison metric defined in this paper. It compares the per-layer profile of a decoder's predictive path using task calibration and DTW with a reported warp, threshold and behavioural check. It is a metric name, not an acronym. The main text retains the descriptive names of its components; Tab. 5 states what each denotes.

Each term denotes one mathematical object defined in Sec. 3 and carries no physical or cognitive claim. Thermodynamic length is path length in the Fisher metric, with no dissipation bound or equilibrium interpretation, since depth is not time; the one inequality that resembles a physical bound, $\mathrm{TL}^2 \le L \sum_k d_k^2$ in Thm. 1, is the Cauchy-Schwarz inequality. "Spectral" refers to the probability spectrum $\sqrt{p}$ and not to eigenvalues. "Belief" is a predictive distribution and not a mental state, and "field" means one tangent vector per layer and not a parameter gradient. The plain aliases (path length, turning curvature, steering field) can be substituted throughout, and no result depends on the names.

C Information geometry: preliminaries and proofs

We collect what the main text uses about the Fisher-Rao geometry of categorical distributions, with proofs. The Fisher metric is the unique Riemannian metric on statistical manifolds that is invariant under sufficient statistics (Chentsov, 1982; Amari, 2016; Ay et al., 2017; Nielsen, 2020); for categorical laws everything is explicit.

C.1 The categorical model and its Fisher metric

The categorical model on $\{1, \dots, n\}$ is the open simplex $\mathring{\Delta}^{n-1} = \{p : p_i > 0,\ \sum_i p_i = 1\}$, parametrised by $\xi = (p_1, \dots, p_{n-1})$ with $p_n = 1 - \sum_{i<n} \xi^i$. With $\ell(\xi; x) = \log p_x$, the Fisher matrix is $g_{ij} = \mathbb{E}[\partial_i \ell\, \partial_j \ell]$.

Proposition 1 (Fisher matrix). *$g_{ij}(\xi) = \delta_{ij}/p_i + 1/p_n$ for $1 \le i, j \le n-1$.*

Proof. $\partial_i \ell(\xi; x) = (\mathbf{1}[x{=}i] - \mathbf{1}[x{=}n])/p_x$. Therefore

$$g_{ij} = \sum_x p_x\, \partial_i \ell\, \partial_j \ell = \frac{\delta_{ij}}{p_i} + \frac{1}{p_n}. \qquad \square$$

Table 5: **What each name means here.** The last two columns are the contract: what the term keeps from its source and what it does not claim.

Term and symbol	Plain alias	Defined here as	Source of the word	Kept	Not claimed
nDNA	checkpoint metric	task-calibrated comparison of length and projected-field profiles; curvature is a secondary descriptor	coined metric name introduced here	a depth-resolved comparison of checkpoints	a universal distance, quality score or causal test
Thermodynamic length $\mathcal{L}_\ell$	path length	$\mathcal{L}_\ell = \mathbb{E}[d_{\mathrm{FR}}(p_\ell, p_{\ell+1})]$: the Fisher-Rao length of the depth path, additive over layers	Weinhold; Salamon and Berry; Crooks: length of a path of distributions in the Fisher metric	an additive length that does not depend on the parametrisation	work, dissipation or equilibrium; depth is not time
Spectral curvature κ_ℓ	turning curvature	$\kappa_\ell = \mathbb{E}[\phi/(a+b)]$: turning angle per unit arc of the belief path	"spectrum": the vocabulary probability vector (amplitudes $\sqrt{p}$); discrete curvature as turning angle	geodesic curvature of the path (Thm. 4)	an eigenspectrum or operator spectrum
Belief vector field $\mathbf{v}_\ell(c)$	correction field	$\mathbf{v}_\ell(c) = \mathbb{E}[\mathrm{PT}(w)]$: the mean transported push toward the target token under condition c	"belief": a predictive distribution; "field": one tangent vector per layer	the mean direction in which a condition raises the target probability	a mental state; a parameter gradient

Proposition 2 (Covariance)**.** *If $\xi = \xi(\theta)$ is a diffeomorphic reparametrisation, the Fisher matrix transforms as $\tilde{G}(\theta) = J^\top G(\xi(\theta))J$ with $J = \partial\xi/\partial\theta$. Hence lengths of curves, and every quantity defined from them, do not depend on the parametrisation.*

Proof. With $\tilde{\ell}(\theta; x) = \ell(\xi(\theta); x)$ the chain rule gives $\partial_{\theta_a}\tilde{\ell} = \sum_k J_{ka}\partial_{\xi_k}\ell$, so $\tilde{g}_{ab} = \mathbb{E}[\partial_{\theta_a}\tilde{\ell}\,\partial_{\theta_b}\tilde{\ell}] = \sum_{k,r} J_{ka}g_{kr}J_{rb}$. The length $\int\sqrt{\dot{\theta}^\top\tilde{G}\dot{\theta}}\,\mathrm{d}t = \int\sqrt{\dot{\xi}^\top G\dot{\xi}}\,\mathrm{d}t$ is therefore unchanged. □

C.2 The square-root isometry and the closed form

Proposition 3 (Isometry)**.** *The map $f(p) = 2\sqrt{p} = (2\sqrt{p_1}, \ldots, 2\sqrt{p_n})$ is an isometry from $(\mathring{\Delta}^{n-1}, g)$ onto the open positive orthant of the sphere of radius two with the metric induced from $\mathbb{R}^n$.*

Proof. f maps into the sphere since $\sum(2\sqrt{p_i})^2 = 4$. In the coordinates ξ,

$$\partial_i f = \tfrac{1}{\sqrt{p_i}}e_i - \tfrac{1}{\sqrt{p_n}}e_n,$$

so $\langle\partial_i f, \partial_j f\rangle = \delta_{ij}/p_i + 1/p_n = g_{ij}$ by Prop. 1. The ∂_i form a basis of the tangent space, so f preserves the metric, and it is a diffeomorphism onto the orthant. □

Corollary 1 (Fisher-Rao distance)**.** *For p, q in the open simplex,*

$$d_{\mathrm{FR}}(p, q) = 2\arccos\Big(\sum_i \sqrt{p_i q_i}\Big), \qquad (8)$$

which is a metric, at most π, and the minimising geodesic is unique.

Proof. Geodesics of a sphere are great circles. For unit vectors $u = \sqrt{p}$, $v = \sqrt{q}$ with angle $\theta \in [0, \pi/2]$ (their inner product is non-negative), the minimising arc is $\gamma(s) = (\sin((1-s)\theta)\,u + \sin(s\theta)\,v)/\sin\theta$, which has positive coefficients and hence stays in the positive orthant, so it lies in the image of f. Its length on the radius-two sphere is 2θ. It is the Riemannian distance of a metric, hence a metric, and the shortest arc is unique because $\theta < \pi$. □

This is Eq. (2) of the main text. As a consequence a path of distributions $p_0, \ldots, p_L$ has a canonical length, the length of the piecewise minimising geodesic through the points, and the triangle inequality holds for it.

C.3 Local relation to the Kullback-Leibler divergence

Remark 3. Let $q = p + \delta$ with $\sum_i \delta_i = 0$, write $x_i = \delta_i / p_i$, $\varepsilon = \max_i |x_i|$ and $S = \sum_i \delta_i^2 / p_i$. As $\varepsilon \to 0$,

$$D_{\mathrm{KL}}(p\|q) = \tfrac{1}{2} S\,(1 + O(\varepsilon)),$$
$$d_{\mathrm{FR}}(p,q)^2 = S\,(1 + O(\varepsilon)),$$

with absolute constants, independent of n and of $\min_i p_i$. Hence $d_{\mathrm{FR}}^2 = 2D_{\mathrm{KL}}\,(1 + O(\varepsilon))$: a short Fisher-Rao step is the square root of twice the local divergence.

Proof. For $|x| \le \frac{1}{2}$, $-\log(1+x) = -x + \frac{1}{2}x^2(1 + \rho(x))$ with $|\rho(x)| \le 2|x|$. Hence $D_{\mathrm{KL}}(p\|q) = -\sum_i p_i \log(1 + x_i) = -\sum_i \delta_i + \frac{1}{2}\sum_i p_i x_i^2 (1 + \rho(x_i))$; the first sum is zero and $\sum_i p_i x_i^2 = S$. For the distance put $r = \|\sqrt{p} - \sqrt{q}\|$. Both vectors are unit, so $\sum_i \sqrt{p_i q_i} = 1 - r^2/2$ and $d_{\mathrm{FR}} = 2\arccos(1 - r^2/2) = 4\arcsin(r/2) = 2r(1 + O(r^2))$. Termwise, $r^2 = \sum_i \delta_i^2 / (\sqrt{p_i} + \sqrt{q_i})^2 = \sum_i \frac{\delta_i^2}{p_i(1+\sqrt{1+x_i})^2} = \frac{1}{4} S\,(1 + O(\varepsilon))$, and $S = \sum_i p_i x_i^2 \le \varepsilon^2$, so $r^2 = O(\varepsilon^2)$ and $d_{\mathrm{FR}}^2 = 4r^2(1 + O(\varepsilon^2)) = S\,(1 + O(\varepsilon))$. □

The relative step ε, not $\|\delta\|$, is the right smallness condition: in a vocabulary of 10^5 tokens many p_i are below 10^{-8}, and an absolute change of 10^{-6} there is not small. The Kullback-Leibler divergence (Kullback and Leibler, 1951) is neither symmetric nor a metric, which is why a metric is used for path length.

C.4 Latent paths in decoder-only transformers

For a decoder with hidden width d and vocabulary size V, position t of a prefix $x_{\le t}$ has a selected sequence of block-output states $h_t^{(\ell)} \in \mathbb{R}^d$. The evaluated profiles exclude the embedding output and reindex the retained block outputs by ℓ. Applying the output head and a temperature gives $z_t^{(\ell)} \in \mathbb{R}^V$ and $p_{\ell,t} = \mathrm{softmax}(z_t^{(\ell)}/\tau)$, so the depth axis defines a path $\ell \mapsto p_{\ell,t}$ in the open simplex.

Two non-identifiabilities make Euclidean geometry in this space unreliable. First, $\mathrm{softmax}(z) = \mathrm{softmax}(z + c\mathbf{1})$, so logits are defined only up to translation. Second, Euclidean distance between probability vectors is not tied to statistical distinguishability: an absolute change of 10^{-3} matters more for the log-likelihood in a low-probability coordinate than in a high-probability one. The Fisher metric weights directions by their effect on the log-likelihood, and by Prop. 2 nothing computed from it depends on the chosen parametrisation. We therefore need step lengths, a turning rate and a steering direction that are intrinsic, statistically meaningful and composable across layers; Apps. D to F supply them.

D Fisher-Rao path length (thermodynamic length)

D.1 Definition and name

Thermodynamic length is the accumulated Fisher-Rao arc length of the depth-indexed predictive path. The term comes from finite-time thermodynamics, where path length in the Fisher metric of a family of distributions was introduced (Weinhold, 1975; Salamon and Berry, 1983; Crooks, 2007). We keep the geometry and drop the physics: transformer depth is not time, the layerwise distributions are not equilibrium states, and no dissipation bound is claimed. A plain alias is *path length*.

D.2 Theorem and proof

Fix a position (x, t) and write $p_\ell = p_{\ell,t}$, $u_\ell = \sqrt{p_\ell}$ and $d_\ell = d_{\mathrm{FR}}(p_\ell, p_{\ell+1}) = 2\arccos\langle u_\ell, u_{\ell+1}\rangle$. The depth is discrete, so the canonical discretisation of the curve length is the length of the piecewise minimising-geodesic interpolation,

$$\mathrm{TL}(x,t) = \sum_{\ell=0}^{L-1} d_\ell, \qquad \mathcal{L}_\ell = \mathbb{E}_{(x,t)\sim P}[d_\ell], \quad (9)$$
$$\mathrm{TL}(P) = \sum_\ell \mathcal{L}_\ell. \quad (10)$$

By Prop. 3 the map $p \mapsto 2\sqrt{p}$ is an isometry onto the open positive orthant of the radius-two sphere, so d_ℓ is twice the angle between u_ℓ and $u_{\ell+1}$, and TL is invariant under any smooth reparametrisation of the categorical model (Prop. 2). All points lie in the open positive orthant, so the spherical distance between any two of them is at most $\pi/2$ and the minimising geodesic is unique.

Proof of Thm. 1. (i) d_{FR} is the distance of a Riemannian metric, hence satisfies the triangle inequality, and applying it along $p_0, \dots, p_L$ gives $d_{\mathrm{FR}}(p_0, p_L) \le \sum_k d_k$. If equality holds, the concatenation of the minimising arcs is a curve whose length equals the distance of its endpoints, so it is a (reparametrised) minimising geodesic; the minimiser is unique, hence the p_k lie on it in order. The

converse is immediate. (ii) The same argument applied to $p_\ell, \dots, p_L$ gives $d_{\mathrm{FR}}(p_\ell, p_L) \le \sum_{k \ge \ell} d_k$, and expectation preserves the inequality. The converse fails: if layer ℓ raises the probability of a token and layer $\ell+1$ lowers it back, both d_ℓ and $d_{\ell+1}$ are large and $p_{\ell+2} = p_\ell$. Length is therefore an upper bound on the remaining prediction distance, not on a layer's causal influence. (iii) The Cauchy-Schwarz inequality gives $(\sum_k d_k)^2 \le L \sum_k d_k^2$ with equality iff all d_k are equal. For the local statement apply Thm. 3 to each pair (p_k, p_{k+1}) with relative step ε_k: $d_k^2 = 2D_{\mathrm{KL}}(p_k \| p_{k+1})(1 + O(\varepsilon_k))$ with an absolute constant. All terms are non-negative, so summing gives $\sum_k d_k^2 = 2\sum_k D_{\mathrm{KL}}(p_k \| p_{k+1})(1 + O(\max_k \varepsilon_k))$, and therefore $\mathrm{TL}^2 \le 2L \sum_k D_{\mathrm{KL}}(p_k \| p_{k+1})(1 + o(1))$. □

Scope of the claim. We claim that (a) $\mathcal{L}_\ell$ is a well-defined, additive, reparametrisation-invariant path length in a local-divergence geometry; (b) it bounds how far later layers can move the prediction (Thm. 1(ii)); and (c) it is decoder-associated in the evaluated cohort (App. J). We do not claim that large length implies influence, or that the geometry has thermodynamic content. A *detour index* $\mathrm{TL}/d_{\mathrm{FR}}(p_0, p_L) \ge 1$ quantifies how much of the length is revision rather than progress; we define it but do not report it because the archived exports do not store $d_{\mathrm{FR}}(p_0, p_L)$.

Scope of use. The remaining length bounds how much later layers moved the output, so it can be used offline to analyse early-exit decoding (Schuster et al., 2022); it is not an online certificate, because computing it requires the remaining layers. Applied to row-stochastic attention kernels instead of vocabulary distributions the same construction gives an attention-transport length, which we evaluated and did not adopt (App. U). In the retained absolute profiles, 41 to 49% of the length of each decoder lies in the first third of depth (Tab. 6).

D.3 Numerical conventions and cost

Time. With B the batch size, S the sequence length, m nodes, H hidden width, V vocabulary and $N = |\mathcal{T}|$ retained positions, one batch costs the forward pass $T_{\mathrm{fwd}}(B, S, m, H)$ plus the readout $\mathcal{O}(mNHV)$ (logit lens), plus $\mathcal{O}(mNV)$ for softmax, square roots, inner products and arccosines. For realistic vocabularies the dense head dominates and the overhead is $\Theta(mNHV)$. Keeping the last K positions gives $N \le BK$ and cuts the cost linearly.

Algorithm 1 Thermodynamic length

Initialise sums $S[\ell] \leftarrow 0$, counts $C[\ell] \leftarrow 0$ for $\ell = 0, \dots, m-2$
for each batch **do**
  Select supervised positions $\mathcal{T} = \{(b,t) : y_{b,t} \ne -100\}$, optionally the last K per sequence
  One forward pass with hidden states; for node $j = 0$: $q_0 \leftarrow \mathrm{softmax}(\mathrm{Lens}(h_0)/\tau)$, $u_0 \leftarrow \sqrt{q_0}$
  for $j = 1$ to $m-1$ **do**
    $q_j \leftarrow \mathrm{softmax}(\mathrm{Lens}(h_j)/\tau)$, $u_j \leftarrow \sqrt{q_j}$ for all positions in $\mathcal{T}$
    $d \leftarrow 2\arccos(\mathrm{clip}(\langle u_{j-1}, u_j \rangle, -1, 1))$; $S[j-1]$ += $\sum d$; $C[j-1]$ += $|\mathcal{T}|$
  end for
end for
Output $\mathcal{L}[\ell] = S[\ell]/\max(C[\ell], 1)$ and $\mathrm{TL}(P) = \sum_\ell \mathcal{L}[\ell]$

Space. Streaming two consecutive nodes needs $\mathcal{O}(NV)$ working memory on top of the stored hidden states $\mathcal{O}(mBSH)$. Materialising distributions for all BS positions instead costs $\mathcal{O}(BSV)$ per node, which is the real bottleneck for large vocabularies; gathering the supervised subset before the readout avoids it. Concrete numbers are in Tabs. 11 and 12.

E Supervised correction field (belief vector field)

E.1 Definition and name

Length and curvature are unsigned: they describe how far and how sharply the prediction moves, not toward what. The belief vector field adds direction. For a supervised token, it takes the tangent direction at the current belief that raises the probability of the target, moves these directions from different points to one reference without twisting or rescaling them, and averages. "Belief" means a predictive distribution, not a mental state, and "field" is literal: one tangent vector per layer. A plain alias is *steering field*.

E.2 Theorem and proof

Let $z = z_{\ell,t}$, $q = \mathrm{softmax}(z/\tau)$, $u = \sqrt{q}$, $\Pi = I - uu^\top$, and $g = \nabla_z \log q_y = \tau^{-1}(e_y - q)$.

Table 6: **Where absolute profile mass occurs** (32 prompts, τ=1). Total length and mean projected-field norm are descriptive within a model; the final column gives the percentage of length in the first, middle and final thirds of depth.

Model	$\sum \mathcal{L}$	$\overline{\|\widetilde{\mathbf{v}}\|}$	depth thirds (%)
Gemma-3 1B	25.48	0.0233	49.3 / 29.6 / 21.0
Llama-3.1 8B	31.29	0.0111	45.9 / 33.8 / 20.2
Qwen-3 4B	38.09	0.0168	40.7 / 34.3 / 24.9
Qwen-3 8B	40.00	0.0203	44.9 / 27.7 / 27.4

Proposition 4 (Logit gradient)**.** $g = \tau^{-1}(e_y - q)$.

Proof. With $Z = \sum_k e^{z_k/\tau}$, $\log q_y = z_y/\tau - \log Z$ and $\partial_{z_j} \log Z = \tau^{-1} q_j$, so $\partial_{z_j} \log q_y = \tau^{-1}(\delta_{jy} - q_j)$. □

Proposition 5 (Pushforward)**.** *The push of a logit step* $\mathrm{d}z$ *through* $z \mapsto \sqrt{\mathrm{softmax}(z/\tau)}$ *is* $\mathrm{d}u = \frac{1}{2\tau}\Pi\,\mathrm{Diag}(u)\mathrm{d}z$*, which is tangent to the sphere.*

Proof. $J = \partial q/\partial z = \tau^{-1}(\mathrm{Diag}(q) - qq^\top)$ and $\mathrm{d}u_i = \mathrm{d}q_i/(2\sqrt{q_i})$, so $\mathrm{d}u = \frac{1}{2\tau}(\mathrm{Diag}(u) - uq^\top)\mathrm{d}z = \frac{1}{2\tau}\Pi\,\mathrm{Diag}(u)\mathrm{d}z$ using $uq^\top = uu^\top\,\mathrm{Diag}(u)$. Tangency: $u^\top \mathrm{d}u \propto q^\top \mathrm{d}z - q^\top \mathrm{d}z = 0$. □

We take $\mathrm{d}z = g$ and write $w = \frac{1}{2\tau}\Pi\,\mathrm{Diag}(u)g$ for the belief push. Temperature enters twice, through g and through the push, which is why absolute field values depend strongly on τ.

Proof of Thm. 2(i). Tangency is Prop. 5. Invariance to a logit shift $z \mapsto z + c\mathbf{1}$ holds because q is unchanged. Write $r = \tau g = e_y - q$, so that componentwise $w_i = \frac{1}{2\tau^2} u_i (r_i - q^\top r)$. Then

$$\begin{aligned}\|w\|^2 &= \tfrac{1}{4\tau^4} \textstyle\sum_i q_i (r_i - q^\top r)^2 \\ &= \tfrac{1}{4\tau^4}\,\mathrm{Var}_{i\sim q}(r_i).\end{aligned}$$

With $r_i = \mathbf{1}[i{=}y] - q_i$: $\mathbb{E}_q[r] = q_y - \sum q_i^2$ and $\mathbb{E}_q[r^2] = q_y(1 - 2q_y) + \sum q_i^3$, which gives the stated closed form (we verified it numerically against the matrix expression to machine precision). For an interior q (and $V \geq 2$) the variance is positive: every $q_i > 0$, so it vanishes only if r is constant, i.e. $1 - q_y = r_y = r_i = -q_i$ for some $i \neq y$; the left side is positive and the right side negative. As q approaches any vertex e_j the variance of r under q tends to zero, whether or not $j = y$: with $1 - q_j = \epsilon$ it is $O(\epsilon^3)$ when $j = y$ and $O(\epsilon)$ when $j \neq y$. Finally $r_i \in [-1, 1]$, so its variance is at most 1 and $\|w\| \leq 1/(2\tau^2)$. □

The norm is therefore not a difficulty score. It is small when the prediction is confident, right or wrong (faster when right), and largest for spread-out predictions; a small *mean* push can also come from individual pushes that cancel, which the coherence $\|\mathbf{v}_\ell\|/\mathbb{E}\|w\|$ separates. The retained coherence summary is reported in Tab. 29.

Common reference. Pushes at different $u_{\ell,t}$ lie in different tangent spaces and cannot be averaged until moved to one. Let $\bar{u}_\ell = \mathbb{E}[u_{\ell,t}]/\|\mathbb{E}[u_{\ell,t}]\|$. Softmax beliefs lie in the positive orthant, so $\langle u_{\ell,t}, \bar{u}_\ell \rangle > 0$ and the short geodesic is unique.

Proposition 6 (Parallel transport)**.** *For unit vectors* u, a *with* $u \neq -a$ *and* $v \in T_u\mathbb{S}^{V-1}$*, transport along the short great-circle arc is* $\mathrm{PT}_{u\to a}(v) = v - \frac{\langle v,a\rangle}{1+\langle u,a\rangle}(u+a)$*. It lies in* $T_a\mathbb{S}^{V-1}$ *and preserves inner products and norms.*

Proof. With $c = \langle u, a \rangle$ decompose the ambient space into the plane $\mathrm{span}\{u, a\}$ and its orthogonal complement. On the complement the map is the identity. In the plane the unit tangent at u toward a is $(a - cu)/\sqrt{1 - c^2}$ and the transported tangent at a is $(ca - u)/\sqrt{1 - c^2}$; substituting either into the formula reproduces this rotation. This is the Levi-Civita transport: along the great circle $\gamma(s) = \cos s\, u + \sin s\, t$ with $t = (a - cu)/\sqrt{1 - c^2}$, the velocity γ' is parallel because γ is a geodesic, and a constant ambient vector orthogonal to $\mathrm{span}\{u, a\}$ stays tangent with zero derivative, hence is parallel; the formula agrees with transport on both, and transport is linear. Directly, $\langle \mathrm{PT}(v), a \rangle = \langle v, a \rangle - \langle v, a \rangle (c + 1)/(1 + c) = 0$. A rotation in one plane and the identity on its complement is orthogonal. □

Proof of Thm. 2(ii). $\|\mathbf{v}_\ell\| = \|\mathbb{E}[\mathrm{PT}\, w]\| \leq \mathbb{E}\|\mathrm{PT}\, w\| = \mathbb{E}\|w\|$ by the triangle inequality and Prop. 6. Equality holds iff all vectors are non-negative multiples of a common vector (equality in Jensen's inequality for the norm). □

The retained empirical runs did not transport each push. They averaged w in $\mathbb{R}^V$ and projected the mean onto $T_{\bar{u}}\mathbb{S}^{V-1}$. The next result shows that

this estimator is close to Eq. (6) when the beliefs of a layer are concentrated, and obeys the same bound.

Proposition 7 (Projected mean)**.** *Let $u, \bar{u}$ be unit vectors with $c = \langle u, \bar{u}\rangle > -1$, $w \in T_u\mathbb{S}^{V-1}$ and $\Pi_{\bar{u}} = I - \bar{u}\bar{u}^\top$. Then*

$$\|\mathrm{PT}_{u\to\bar{u}}(w) - \Pi_{\bar{u}}w\| \le \tfrac{1}{2}\|w\|\,\|u-\bar{u}\|^2,$$

with equality when w points along the great circle through u and $\bar{u}$. Consequently $\widetilde{\mathbf{v}}_\ell = \Pi_{\bar{u}_\ell}\mathbb{E}[w_{\ell,t}]$ satisfies $\|\widetilde{\mathbf{v}}_\ell - \mathbf{v}_\ell\| \le \frac{1}{2}\mathbb{E}[\|w_{\ell,t}\|\|u_{\ell,t}-\bar{u}_\ell\|^2]$ and $\|\widetilde{\mathbf{v}}_\ell\| \le \mathbb{E}\|w_{\ell,t}\|$.

Proof. From Prop. 6, $\mathrm{PT}(w) - \Pi_{\bar{u}}w = \langle w,\bar{u}\rangle\big(\bar{u} - \frac{u+\bar{u}}{1+c}\big) = \langle w,\bar{u}\rangle\frac{c\bar{u}-u}{1+c}$. Since $\langle w,u\rangle = 0$, $\langle w,\bar{u}\rangle = \langle w, \bar{u}-cu\rangle$, and $\|\bar{u}-cu\| = \|c\bar{u}-u\| = \sqrt{1-c^2}$. By Cauchy-Schwarz the difference has norm at most $\|w\|(1-c^2)/(1+c) = \|w\|(1-c) = \frac{1}{2}\|w\|\|u-\bar{u}\|^2$, with equality when $w \parallel \bar{u} - cu$. The second claim follows because $\Pi_{\bar{u}}$ is linear, so $\widetilde{\mathbf{v}}_\ell = \mathbb{E}[\Pi_{\bar{u}_\ell}w_{\ell,t}]$, and from the triangle inequality. The third holds because $\Pi_{\bar{u}}$ is a contraction: $\|\widetilde{\mathbf{v}}_\ell\| \le \|\mathbb{E}w\| \le \mathbb{E}\|w\|$. □

The *coherence* $\|\mathbf{v}_\ell\|/\mathbb{E}\|w\| \in [0,1]$ separates weak pushes from strong but mutually inconsistent ones, which a small norm alone cannot do. The projected ambient mean used in the archived runs differs from the transported mean by a term of second order in the spread of the beliefs around $\bar{u}_\ell$ (Prop. 7).

Scope of use. The field depends on the prompt distribution chosen to represent a condition, so it is the channel that follows the task (Tab. 2); this matters when a condition is the object of study and is a nuisance when only the decoder is. Reporting it separately keeps task effects from being attributed to the decoder (App. J). The cosine between fields could localise where two conditions agree or conflict; that directional comparison is not performed here.

E.3 Numerical conventions and cost

Time and space. No backward pass is needed: g and w have closed forms. The exact estimator uses two passes, so its time is $2T_{\text{fwd}} + \mathcal{O}(LNHV) + \mathcal{O}(LNV)$ per batch; the readouts dominate and the push and transport are linear in LNV. Peak working memory is $\mathcal{O}(NV)+\mathcal{O}(LV)$ when distributions are streamed by layer (the anchors $\bar{u}_\ell$ take $\mathcal{O}(LV)$). A one-pass variant caches

Algorithm 2 Parallel-transported belief vector field

Pass 1: accumulate $U[\ell] \leftarrow \sum_{(x,t)} u_{\ell,t}$ and counts; set $\bar{u}_\ell \leftarrow U[\ell]/\|U[\ell]\|$
for each batch of the condition c (Pass 2) **do**
  for $\ell = 0$ to $L-1$ **do**
    $q \leftarrow \mathrm{softmax}(\mathrm{Lens}(h_\ell)/\tau)$, $u \leftarrow \sqrt{q}$, $g \leftarrow \tau^{-1}(e_y - q)$, $s \leftarrow q^\top g$
    $w \leftarrow (2\tau)^{-1}(u \odot g - u \odot s)$ (rowwise)
    $\tilde{w} \leftarrow w - \frac{\langle w,\bar{u}_\ell\rangle}{1+\langle u,\bar{u}_\ell\rangle}(u+\bar{u}_\ell)$; accumulate $V[\ell] \mathrel{+}= \sum \tilde{w}$
  end for
end for
Output $\mathbf{v}_\ell = V[\ell]/C[\ell]$, its norm and the dispersion of the transported vectors

$\mathcal{O}(LNV)$ pushes. A backward pass is required only for the alternative *final-output sensitivity* field $\nabla_{h^{(\ell)}} \log p^{(L)}(y^\star \mid x)$, which we do not use for the reported profiles.

F Prediction-path turning curvature (spectral curvature)

F.1 Definition and name

Curvature asks whether the next layer continues the current revision or redirects it, measured as turning per unit of Fisher-Rao arc length. "Spectral" refers to the vocabulary probability spectrum (the vector $\sqrt{p}$ of amplitudes), not to an eigendecomposition; a plain alias is *turning curvature*. Discrete turning angle has been used as a measure of trajectory straightness in vision (Hénaff et al., 2019) and in language models (Hosseini and Fedorenko, 2023); here it is applied to the path of predictions and normalised by arc length.

F.2 Theorem and proof

Let $u_\ell = \sqrt{p_\ell}$ and $a = \arccos\langle u_{\ell-1}, u_\ell\rangle$, $b = \arccos\langle u_\ell, u_{\ell+1}\rangle$, $c = \arccos\langle u_{\ell-1}, u_{\ell+1}\rangle$. All inner products are non-negative because beliefs lie in the positive orthant, so $a, b, c \in [0, \pi/2]$ and $a + b \le \pi$. When $\sin a \sin b > 0$ the spherical law of cosines gives

$$\cos\alpha = \frac{\cos c - \cos a\cos b}{\sin a \sin b}, \tag{11}$$

$$\phi = \pi - \alpha, \qquad \kappa = \frac{\phi}{a+b}. \tag{12}$$

The mean of the two adjacent Fisher-Rao step lengths is $\frac{1}{2}(2a+2b) = a+b$, so κ divides the

turn by the arc length over which it occurs, in Fisher-Rao units. This is the mathematical estimator used in Thm. 4. The numerical algorithm may add the machine epsilon of the arithmetic dtype after excluding degenerate triples; that safeguard is not part of the limiting statement. The layer profile is $\kappa_\ell = \mathbb{E}[\kappa_{\ell,t} \mid \sin a \sin b > \delta]$, where $\delta > 0$ is a declared floating-point exclusion tolerance. The known-curvature simulations contain no degenerate triples and use no exclusion; corrected-curvature model runs must report both δ and the retained count per layer.

Theorem 4 (Curvature). *(i) $\phi \in [0, \pi]$, and $\phi = 0$ iff $c = a + b$, i.e. iff the three beliefs lie in this order on one Fisher-Rao geodesic.*

(ii) $\kappa_{\ell,t}$ depends only on Fisher-Rao distances, hence not on the parametrisation of the categorical model.

(iii) If $p_{\ell-1,t}, p_{\ell,t}, p_{\ell+1,t} = \gamma(s - h_1), \gamma(s), \gamma(s + h_2)$ for a C^2 curve γ in the open simplex parametrised by Fisher-Rao arc length, then $\kappa_{\ell,t} \to k(s)$, the Fisher-Rao geodesic curvature of γ at s, as $h_1, h_2 \to 0$.

Proof of Thm. 4(i)-(ii). $\alpha \in [0, \pi]$ so $\phi \in [0, \pi]$. If $c = a + b$ then $\cos c = \cos a \cos b - \sin a \sin b$, so $\cos \alpha = -1$, $\alpha = \pi$, $\phi = 0$. Conversely $\phi = 0$ gives $\cos c = \cos(a + b)$; since $c \in [0, \pi/2]$ and $a+b \in [0, \pi]$ with cos injective on $[0, \pi]$, $c = a+b$, which holds iff the three points lie in order on one great-circle arc, i.e. on one minimising Fisher-Rao geodesic. The side lengths are (half) Fisher-Rao distances, hence invariant under isometries, and α is a function of them alone. □

Proof of Thm. 4(iii). Work on the unit sphere $\mathbb{S}$ with the curve $\tilde{\gamma} = \sqrt{\gamma}$ parametrised by spherical arc length σ, centred so that $\tilde{\gamma}(0) = u_{\ell,t} =: x$; the neighbours are $\tilde{\gamma}(-s_1)$ and $\tilde{\gamma}(s_2)$ with $s_1, s_2 > 0$. In normal coordinates at x, define $y(\sigma) = \exp_x^{-1}(\tilde{\gamma}(\sigma)) \in T_x\mathbb{S}$. The metric at the origin is Euclidean, the Christoffel symbols vanish there, and radial geodesics are straight lines. Since $\tilde{\gamma}$ is C^2,

$$y(\sigma) = \sigma v + \tfrac{1}{2}\sigma^2 A + o(\sigma^2),$$

where v is the unit tangent and $A = \nabla_{\tilde{\gamma}'}\tilde{\gamma}'(0) \perp v$ is the acceleration, whose norm is the geodesic curvature $k^{\mathbb{S}}$ (possibly 0). The radial geodesic from x to $\tilde{\gamma}(\sigma)$ has normal-coordinate length $\|y(\sigma)\| = |\sigma|(1 + o(1))$ and initial unit direction $y(\sigma)/\|y(\sigma)\| = \operatorname{sgn}(\sigma)\big(v + \tfrac{1}{2}\sigma A\big) + o(\sigma)$, using $A \perp v$. Hence $a = s_1(1 + o(1))$, $b = s_2(1 + o(1))$, the direction to the next point is $e_+ = v + \frac{1}{2}s_2 A + o(s_2)$ and minus the direction to the previous point is $e_- = v - \frac{1}{2}s_1 A + o(s_1)$. The law of cosines on $\mathbb{S}$ returns the Riemannian angle α at x between the two geodesic sides, which in these coordinates is the Euclidean angle between e_+ and $-e_-$; the exterior turn $\phi = \pi - \alpha$ is therefore the angle between e_+ and e_-, namely $\|e_+ - e_-\|(1 + o(1)) = \frac{1}{2}k^{\mathbb{S}}(s_1 + s_2) + o(s_1 + s_2)$. Dividing, $\phi/(a + b) \to \frac{1}{2}k^{\mathbb{S}}$; this also covers $k^{\mathbb{S}} = 0$, where $\phi = o(s_1 + s_2)$. Finally the Fisher-Rao metric is 4 times the round metric with the same Levi-Civita connection, so Fisher-Rao arc length is $s = 2\sigma$ (thus $h_i = 2s_i$) and $\nabla_{d/ds}\frac{d\gamma}{ds} = \frac{1}{4}A$, whose Fisher-Rao norm is $2 \cdot \frac{1}{4}\|A\| = \frac{1}{2}k^{\mathbb{S}} = k(s)$. Hence $\kappa_{\ell,t} \to k(s)$. For small h_1, h_2 we have $a, b > 0$ and $a + b < \pi$, so the triple is non-degenerate and the formula applies. □

Table 7: **Curvature estimators on 40 simulated paths.** Error and correlations compare the corrected and original estimators with known Fisher–Rao geodesic curvature.

L	η	error	r (corrected)	r (original)	r (original, inverse step)
16	0.0	0.2%	1.00	-0.66	1.000
16	0.1	71.1%	-0.26	-0.67	0.999
16	0.3	150.5%	-0.42	-0.63	0.994
32	0.0	0.1%	1.00	-0.66	1.000
32	0.1	137.1%	-0.42	-0.65	0.999
32	0.3	485.0%	-0.43	-0.58	0.991
64	0.0	0.0%	1.00	-0.65	1.000
64	0.1	307.1%	-0.37	-0.66	0.999
64	0.3	1005.7%	-0.43	-0.62	0.987

Numerical check. On a small circle of colatitude 0.9 on $\mathbb{S}^2$ (geodesic curvature $\cot 0.9$ per spherical arc, so $\frac{1}{2}\cot 0.9 = 0.39678$ per Fisher-Rao arc), the estimator returns 0.39852, 0.39688, 0.39678 at spacings 0.2, 0.05, 0.01.

We also evaluate 40 smooth positive-orthant paths with known, varying geodesic curvature and front-loaded node spacing (Tab. 7). Without perturbation, median relative error is 0.22% at 16 nodes and 0.01% at 64 nodes. Perturbing nodes by one tenth of an adjacent step makes the estimator unstable, which motivates float32 readouts and explicit exclusion of degenerate triples for model-scale use.

Degenerate triples. The formula is ill-conditioned when $\sin a \sin b \approx 0$ (a stalled step or antipodal points). Such triples carry no directional information, so they are *excluded* from the expectation (threshold δ) and not counted as zero, which would bias κ_ℓ downward exactly

Table 8: **Original curvature-channel diagnostics**, 32 prompts, τ=1. Mean and range of the interior-angle score, its Pearson correlation with the chord estimator, and with the inverse step size $\pi/(a+b)$ across layers.

Model	mean s_{int}	range	r (chord)	r ($\pi/(a+b)$)
Gemma-3 1B	3.855	[0.60, 9.57]	-0.303	0.859
Llama-3.1 8B	2.539	[0.93, 8.42]	0.609	0.902
Qwen-3 4B	2.538	[1.03, 8.78]	0.423	0.891
Qwen-3 8B	2.171	[0.78, 5.46]	-0.073	0.801

where the path stalls. The original runs retained between 208 and 246 of 256 positions at the least populated layer, so the exclusion rate should be plotted beside κ.

F.3 The original interior-angle implementation

The original configuration is labelled "turn" but normalises the interior angle α instead of the exterior turn ϕ. Because $\alpha = \pi - \phi$,

$$s_{\text{int}} = \frac{\alpha}{a+b+\varepsilon} = \frac{\pi}{a+b+\varepsilon} - \kappa,$$

which is close to $\pi/(a+b)$ on a nearly straight path: a score dominated by the inverse step size. This is what the data show: across layers, s_{int} correlates 0.80 to 0.90 with $\pi/(a+b)$ (Tab. 8). A second estimator used in those runs, the projected second difference $\|\Pi\Delta^2 u\|/(\|\Delta u\|^2+\varepsilon)^{3/2}$, is extrinsic and uses chord rather than arc length; it agrees with neither (Pearson r from -0.30 to $+0.61$). Length and field do not use α and are unaffected. All curvature values produced by the original runs are therefore interior-angle scores rather than corrected curvature. Every dashboard result in this paper is computed from length and field; the stored third channel appears only in rows labelled "original angle," and the four-model exports (task threshold and behaviour block) retain it. Corrected curvature is validated by the theorem, known-curvature simulation and the matched aggregate ablation in Tab. 26. The retained per-layer dashboards cannot be retroactively converted from the interior angle because they do not store the required position-level triples.

F.4 Numerical conventions and cost

Time and space. Three inner products and a few trigonometric operations per position and node add $\mathcal{O}(mNV)$ to the readout cost $\mathcal{O}(mNHV)$. Three consecutive embeddings (u_{r-1}, u_r, u_{r+1}) are needed at once, so working memory is $\mathcal{O}(NV)$ plus the stored hidden states $\mathcal{O}(mBSH)$, and the accumulators are $\mathcal{O}(m)$. Keeping the last K positions again gives $N \le BK$.

Algorithm 3 Spectral curvature (exterior turn)

Initialise $S[r] \leftarrow 0$, $C[r] \leftarrow 0$ for interior nodes $r = 1, \dots, m-2$
for each batch **do**
  One forward pass; for each node $q \leftarrow \text{softmax}(\text{Lens}(h)/\tau)$, $u \leftarrow \sqrt{q}$ at supervised positions
  for $r = 1$ to $m-2$ **do**
    $a \leftarrow \arccos\langle u_{r-1}, u_r\rangle$, $b \leftarrow \arccos\langle u_r, u_{r+1}\rangle$
    $c \leftarrow \arccos\langle u_{r-1}, u_{r+1}\rangle$ (all clipped)
    if $\sin a \sin b > \delta$ **then**
      $\alpha \leftarrow \arccos\big((\cos c - \cos a \cos b)/(\sin a \sin b)\big)$; $\phi \leftarrow \pi - \alpha$
      $S[r] \mathrel{+}= \phi/(a+b+\varepsilon_{\text{mach}})$; $C[r] \mathrel{+}= 1$
    end if
  end for
end for
Output $\kappa[r] = S[r]/\max(C[r], 1)$ and the number of retained positions $C[r]$

Table 9: **Seven curve distances on 19 models: related-checkpoint retrieval** using the two-channel (original angle, $\mathcal{L}$) matrices retained in the report (32 resampled depth points). NN counts models whose nearest neighbour shares the lineage; AUC measures whether same-lineage pairs are closer; MRR is the reciprocal rank of the first same-lineage model.

Distance	NN	AUC	MRR
PCM	14/19	0.813	0.798
Area between curves	15/19	0.879	0.867
Curve length	14/19	0.786	0.779
Discrete Fréchet	10/19	0.769	0.631
DTW	18/19	0.921	0.955
MAE (resampled)	16/19	0.858	0.884
MSE (resampled)	16/19	0.847	0.878

G Dynamic time warping

G.1 Definition and use

The layer index is ordered but not metrically rigid: the same predictive transition may occur a few blocks earlier in one condition than in another. Dynamic time warping (DTW) compares two multi-

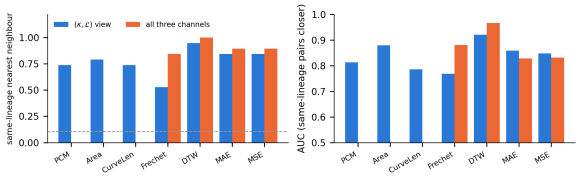
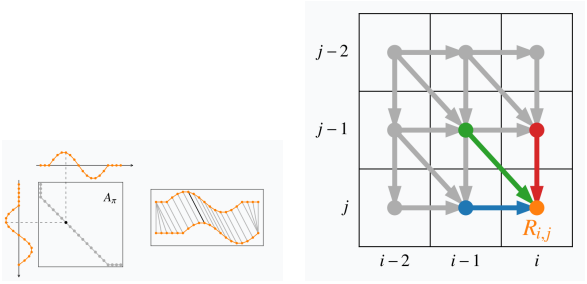


Figure 4: Related-checkpoint retrieval by curve distance (stored report matrices using length and the original interior-angle score). The dashed line is chance (11%).



(a) Minimum-cost path. (b) Permitted transitions.

Figure 5: **How the comparison works.** The path moves monotonically through the layerwise cost matrix and absorbs small shifts in the depth at which a feature appears; the band bounds how far it may leave the diagonal.

channel depth profiles while allowing small local shifts of the depth axis (Sakoe and Chiba, 1978; Itakura, 1975; Müller, 2007). Here it is a validator and not a learned loss, so differentiable relaxations (Cuturi and Blondel, 2017; Abid and Zou, 2018) are unnecessary. It has two uses: to calibrate, because the distance between the task prototypes of one decoder defines the threshold θ, and to compare, because the distance between a base and a modified or related model is read against θ. Both uses require profiles that were scaled together, since costs computed under different scalings are not comparable. An alignment budget is needed for the comparison: without any alignment rule a harmless two-layer shift can look larger than an operation, and without a constraint the alignment could match an early-layer feature to a late-layer one.

Let $\mathbf{z}_i, \mathbf{z}'_j \in \mathbb{R}^d$ stack the channels used (original angle, $\mathcal{L}, \|\mathbf{v}\|$ in the retained task-threshold exports; $(\mathcal{L}, \|\mathbf{v}\|)$ for every dashboard result) after one shared min-max map, $c_{ij} = \|\mathbf{z}_i - \mathbf{z}'_j\|^2$, and $\mathcal{B}$ a Sakoe-Chiba band (half-width 0.2 of the depth; the archived experiments first resample every profile to 32 depth points, the dashboard analyses of this paper use native depths). Define

$$D_{ij} = c_{ij} + \min\{D_{i-1,j}, D_{i,j-1}, D_{i-1,j-1}\}, \quad (i,j) \in \mathcal{B}, \tag{13}$$

with $D_{11} = c_{11}$ and states outside the band equal to $+\infty$.

Proposition 8 (Optimality). *$D_{LL'}$ is the minimum accumulated cost over all monotone, continuous paths from $(1,1)$ to (L,L') inside the band.*

Proof. Every admissible path ending at (i,j) enters it from exactly one of $(i-1,j)$, $(i,j-1)$, $(i-1,j-1)$. If the prefix to that predecessor were not of minimum cost, replacing it by a cheaper admissible prefix would give a cheaper path to (i,j), a contradiction; so an optimal path has optimal prefixes and its cost obeys the recurrence. Induction on $i+j$ from $(1,1)$ shows that every D_{ij} is the optimal cost to its state. The band removes states but leaves the argument unchanged on the remaining admissible paths. □

For analyses recomputed from retained profiles we report $D_{\mathrm{RMS}} = \sqrt{D_{LL'}/|\pi^\star|}$, so the dissimilarity has the units of the scaled profile coordinates and distance-based ANOVA squares it exactly once. Aggregate-only operation tables retain their original mean-squared aligned cost $D_{LL'}/|\pi^\star|$ and are labelled accordingly. For unequal depths, the warping extent is $|\pi^\star|^{-1} \sum_{(i,j)\in\pi^\star} |(i-1)/(L-1) - (j-1)/(L'-1)|$. When several paths attain the minimum, $\pi^\star$ is the one returned by the standard backtrack (diagonal first), so both numbers are well defined. Equal-depth archived tables may additionally report warp in layers or thirty-seconds of depth after resampling. The threshold of a model is $\theta = q_{90}$ of the 45 unique task-pair distances (the stored 10×10 matrix is symmetric, and the pairs share the same ten task profiles). DTW is not a metric: the triangle inequality need not hold, and it does not compare final predictions.

A diagonal move of the path aligns one layer with one layer, and a horizontal or vertical move lets a feature persist for an extra block in one profile; the band bounds how far the path may leave the diagonal (Fig. 5).

G.2 Results

Tab. 14 gives the task variation per model. Tab. 10 adds the mean DTW of each task to the other nine. The most distant task is MNLI for three models and IMDB for Qwen-3 8B; the closest pair involves CNN/DailyMail in three of four models (with WMT16 for Gemma, SQuAD for Llama,

Table 10: **Which tasks widen the within-checkpoint reference.** Each cell is the task's mean DTW to the other nine tasks for that decoder (joint scaling, n=256).

Task	Gemma-3 1B	Llama-3.1 8B	Qwen-3 4B	Qwen-3 8B
ARC-Challenge	0.0250	0.0065	0.0260	0.0304
HellaSwag	0.0211	0.0068	0.0309	0.0232
WinoGrande	0.0358	0.0086	0.0434	0.0247
MNLI	0.0501	0.0169	0.0446	0.0345
QQP	0.0257	0.0069	0.0276	0.0239
SQuAD v2	0.0235	0.0060	0.0345	0.0276
CNN/DailyMail	0.0240	0.0056	0.0278	0.0250
CommonGen	0.0329	0.0109	0.0430	0.0327
WMT16	0.0227	0.0105	0.0426	0.0245
IMDb	0.0231	0.0085	0.0292	0.0370

QQP for Qwen-3 8B). Warping is small: the mean warp over task pairs is 0.8 to 1.6 resampled depth points with a maximum of 3.4, and 4-bit quantisation and siblings stay at or below 1.2 while destroyed models reach 2 to 3.9 (Tab. 40).

G.3 Comparison with other distances

Tab. 9 provides the empirical comparison: DTW performs best among seven distances for related-checkpoint retrieval in the retained two-channel report. The matrices contain the original interior-angle score and are resampled to 32 depth points, so this result supports the alignment rule used in the retained comparison, not corrected curvature or universal superiority over sequence metrics. DTW is appropriate here because the profiles are short ordered curves whose phase may shift by a few layers; pointwise distances penalise such shifts, while discrete Fréchet uses only the worst matched point.

G.4 Scope of interpretation

The small observed warp means that these task profiles can be aligned without large depth remapping. It does not show insensitivity to decoding settings, sequence length or inputs far from the training distribution, and shape comparisons are restricted to models within a factor of two in depth. Task profiles do not coincide, but their observed range is model-specific and smaller than the displacement of the tested destructive operation. DTW is not a metric, and its normalised costs are meaningful only for profiles transformed by the same map. The map used for θ was fitted on task and operation profiles together, so θ is not independent of the operation block; a frozen scaler fitted on task profiles alone is the clean alternative. A low cost does not mean that two models make the same predictions. Every reported cost should state the band, the warp extent, the normalisation scope, the number of unique pairs and their dependence.

G.5 Cost

Without a band the cost matrix and recurrence take $\mathcal{O}(LL'd)$ time for d channels and $\mathcal{O}(LL')$ memory if backpointers are kept; two-row evaluation reduces cost-only memory to $\mathcal{O}(L')$. A band of fixed *fractional* half-width w visits about $2wLL'$ states, so time stays $\mathcal{O}(wLL'd)$, quadratic in depth with a smaller constant; only a band of fixed *integer* width k gives $\mathcal{O}(k \max\{L, L'\}d)$. For the pairwise task matrices in this paper ($L \approx 30$, 45 pairs per model) the cost is negligible next to profiling.

H Computation, tokens and time

Tokens. A profile block uses 256 prompts and at most the last eight loss-bearing positions per prompt, so $N \leq 2{,}048$ supervised positions; padding and special tokens are excluded. The absolute-profile blocks use 32 prompts ($N \leq 256$). Prompt lengths are not stored, so the total number of forward tokens is not reported. Dashboards use 1000 or 2500 prompts per model.

Cost per profile block. Let $m = L + 1$ readout nodes, N retained positions, H the hidden width. After one forward pass, the gathered readouts cost $\mathcal{O}(mNHV)$ and all three descriptors add $\mathcal{O}(mNV)$. Working memory is $\mathcal{O}(NV)$ when two (length, field) or three (curvature) consecutive nodes are streamed, plus the stored hidden states $\mathcal{O}(mNH)$; the exact field needs a second pass for the anchor. The readout dominates, and the archived configuration is small: 256 prompts, at most eight loss-bearing positions per prompt, so $N \leq 2{,}048$ positions per profile block (Tab. 11).

Per-descriptor cost. Tab. 12 lists time and memory. All three descriptors share one forward pass and one set of readouts; the readout $\mathcal{O}(mNHV)$ dominates.

Wall-clock time. Tab. 13 gives the archived run times. The exports record elapsed time but not hardware or peak memory; repository environment files pin PyTorch 2.9.0 and Transformers 4.57.3, without a per-run environment snapshot. The times are therefore not hardware-normalised benchmarks. The absolute-profile runs use 32 prompts, so their times are not comparable with the 256-prompt runs; two of them (Llama-3.1 8B and Qwen-3 8B, 412 and 383 s) are much longer than the others for reasons the exports do not record. Per profile block

Table 11: **Tokens, compute and memory per profile block.** m readout nodes, H hidden width, V vocabulary, N=256 × 8 supervised positions. Lens FLOPs = $2mNHV$; NV is one float32 node; "3 nodes" is the streaming working set; "all nodes" is what a naive implementation would hold. Architecture constants are from public model cards. Wall time is the archived task-stability run divided by its 14 or 15 profile blocks; it includes model loading and the operation variants, so it is an upper bound per block, and the hardware was not recorded.

	m	H	V	N	Lens TFLOPs	NV GB	3 nodes GB	all nodes GB	hidden MB	run s	blocks	s/block
Gemma-3 1B	27	1152	262k	2048	33.4	2.15	6.44	58.0	127	260	14	18.6
Llama-3.1 8B	33	4096	128k	2048	71.0	1.05	3.15	34.7	554	526	14	37.6
Qwen-3 4B	37	2560	152k	2048	58.9	1.24	3.73	46.1	388	652	15	43.4
Qwen-3 8B	37	4096	152k	2048	94.3	1.24	3.73	46.1	621	584	14	41.7

Table 12: **Asymptotic cost per batch.** m nodes, N positions, H width, V vocabulary, B, S batch and sequence length.

Quantity	Time (beyond forward pass)	Working memory
Readouts (shared)	$\mathcal{O}(mNHV)$	$\mathcal{O}(NV)$ streamed
Thermodynamic length	$\mathcal{O}(mNV)$	2 nodes, $\mathcal{O}(NV)$
Spectral curvature	$\mathcal{O}(mNV)$	3 nodes, $\mathcal{O}(NV)$
Belief field (exact)	$\mathcal{O}(mNV)$, two passes	$\mathcal{O}(NV + mV)$
DTW (pair, fractional band w)	$\mathcal{O}(wLL'd)$	$\mathcal{O}(L')$ cost only
Stored hidden states		$\mathcal{O}(mBSH)$

Table 13: **Archived wall-clock time by experiment** (seconds). Runs used different prompt counts, so columns describe computational scale and should not be compared as hardware-normalised benchmarks.

Model	behaviour	task reference	patching	absolute (n=32)
Gemma-3 1B	1758	260	236	11
Llama-3.1 8B	1524	526	529	412
Qwen-3 4B	2342	652	435	24
Qwen-3 8B	2013	584	469	383

the task-stability run averages 19 to 43 seconds (Tab. 11).

Memory in practice. Two stores must be kept apart: the hidden states that a forward pass returns for every position, $\mathcal{O}(mBSH)$, and the readouts for the N gathered positions only, $\mathcal{O}(NV)$ per node. A float32 node of gathered readouts costs $4NV$ bytes: 2.2 GB for Gemma-3 1B and 1.0 to 1.2 GB for the others at $N = 2{,}048$. Holding all nodes at once would need 35 to 58 GB, so the three-node streaming implementation (3 to 6 GB) is what makes the computation feasible on one device; gathering supervised positions before the head is essential for vocabularies above 100k.

I Task reference: subsets of the ten tasks

I.1 Setup

The question is how the task reference and the profile change as tasks are added. If adding tasks kept changing the threshold or a model's identity, the profile would be tied to the task suite. Two experiments address it. (a) The threshold θ_b is recomputed from every subset of b of the ten tasks using the stored distance matrices of the four decoders (three channels, original interior-angle score, joint scaling). (b) For each decoder, the profile averaged over b of the ten tasks is compared with the decoder's ten-task average (dashboard profiles, length and field). Calibration costs ten tasks times the profiling cost per task, so a threshold that stabilises after few tasks makes calibration cheap.

(a) For a subset S with $|S| = b$ the threshold is the 90th percentile of the $\binom{b}{2}$ within-subset DTW distances; we report its median and range over all subsets relative to θ_{10}, the smallest margin of 40% pruning over θ_b, and the fraction of subsets in which the sibling or the 4-bit variant exceeds θ_b. (b) For each decoder and subset we compute the DTW distance from the b-task aggregate to the ten-task aggregate, and ask whether it is nearer to its own ten-task aggregate than to any other decoder's.

I.2 Results

Tab. 15 and Fig. 6: at b=6 the median θ_b/θ_{10} is within 3% of 1 for all four decoders. It is not monotone at seven and eight tasks, although the subset range generally narrows (at b=3 it spans 0.16 to 1.65 times θ_{10}). Prune-40 exceeds the *worst-case* θ_b by 1.9 to 6.4× at b=3 and by 3.2 to 10.5× at b=10. The 4-bit variant never exceeds it; siblings do only for Gemma with few tasks (Tab. 16). Tab. 17: against the ten-task reference, the model is identified from a single task 97.5% of the time, 98.9% from two, and in all evaluated subsets from three on. That reference shares tasks with the subset, so we repeat the test with references built only from the tasks not in the subset: 95% from one task and 97 to 99% from two to seven, falling to 93% and 85% when the reference has only two or one task left; while the distance to the full aggregate

Table 14: **Task reference and the operations read against it** (ten FLAN tasks, 256 prompts; 45 unique task pairs per model; three channels with the original interior-angle score; joint scaling of tasks and variants). θ is the 0.9 quantile of the task-pair distances. Operation columns are distances from the base; bold values exceed θ; the last column is prune-40 divided by θ. Siblings: instruct (all but Qwen-3 8B), then think (Qwen-3 4B); the Qwen-3 8B variant is a base sibling.

	task variation					operations			
Model	min	median	mean	max	θ	quant-4	prune-40	sibling(s)	ratio
Gemma-3 1B	0.0059	0.0267	0.0284	0.0630	0.0524	0.0064	**0.2540**	0.0262	4.9×
Llama-3.1 8B	0.0012	0.0072	0.0087	0.0257	0.0153	0.0010	**0.1602**	0.0012	10.5×
Qwen-3 4B	0.0126	0.0319	0.0350	0.0908	0.0530	0.0120	**0.1702**	0.0098, 0.0133	3.2×
Qwen-3 8B	0.0081	0.0263	0.0283	0.0544	0.0422	0.0086	**0.2524**	0.0106	6.0×

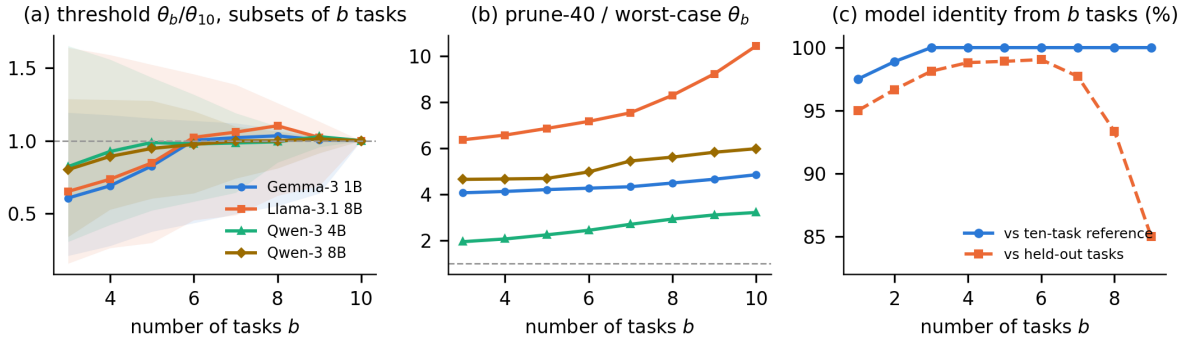


Figure 6: **Task breadth.** (a) The threshold from subsets of b tasks relative to the ten-task threshold (median and range over all subsets). (b) Prune-40 divided by the largest θ_b. (c) The model is identified from the aggregate of b tasks (four models, all subsets).

Table 15: **Breadth of the calibration.** For each decoder: median θ_b/θ_{10} over all subsets of b tasks and the margin prune-40 / $\max_S \theta_b$.

	Gemma-3 1B		Llama-3.1 8B		Qwen-3 4B		Qwen-3 8B	
b	θ_b/θ_{10}	margin	θ_b/θ_{10}	margin	θ_b/θ_{10}	margin	θ_b/θ_{10}	margin
3	0.61	4.1	0.65	6.4	0.82	1.9	0.80	4.7
4	0.69	4.1	0.73	6.6	0.93	2.1	0.89	4.7
5	0.83	4.2	0.85	6.9	0.99	2.2	0.95	4.7
6	1.00	4.3	1.02	7.2	0.98	2.4	0.97	5.0
7	1.02	4.3	1.06	7.5	0.98	2.7	1.00	5.5
8	1.03	4.5	1.10	8.3	0.99	2.9	1.00	5.6
9	1.01	4.7	1.02	9.2	1.03	3.1	1.01	5.8
10	1.00	4.9	1.00	10.5	1.00	3.2	1.00	6.0

Table 16: **Fraction of b-task subsets in which a sibling exceeds θ_b** (the larger of the siblings where there are two). 4-bit quantisation never exceeds θ_b.

b	Gemma-3 1B	Llama-3.1 8B	Qwen-3 4B	Qwen-3 8B
3	25%	0%	0%	0%
5	5%	0%	0%	0%
7	1%	0%	0%	0%
9	0%	0%	0%	0%

falls by a factor of 70 from b=1 to b=9 (the aggregate shares more and more of its tasks with the full one) and the margin over the nearest other decoder rises from 2.2 to 84.

Table 17: **Identity from b tasks** (length and field). Mean over four decoders and all subsets: median and maximum DTW from the b-task aggregate to the ten-task aggregate, share of subsets in which the aggregate is nearest to its own decoder, median margin of the nearest other decoder, and the share identified when every decoder's reference is built only from the $10 - b$ tasks not in the subset (held out).

b	median	max	identified	margin	held out
1	0.1670	0.3748	97.5%	1.5×	95.0%
2	0.0957	0.2887	98.9%	2.2×	96.7%
3	0.0778	0.2375	100.0%	2.6×	98.1%
4	0.0626	0.1904	100.0%	3.2×	98.8%
5	0.0528	0.1646	100.0%	3.8×	98.9%
6	0.0428	0.1317	100.0%	4.6×	99.0%
7	0.0353	0.1018	100.0%	5.5×	97.7%
8	0.0249	0.0740	100.0%	7.9×	93.3%
9	0.0205	0.0449	100.0%	9.1×	85.0%

I.3 Scope of interpretation

In this set, the median six-task threshold is within a few percent of θ_{10}, although the sequence is not monotone at seven and eight tasks; three-task aggregates still identify each of the four decoders.

Table 18: **Leave-one-task-family-out evaluation.** Every task appears in exactly one held-out fold; all preprocessing is fitted on the remaining task families.

Held-out family	train tasks	test tasks	decoder ID	within reference
reasoning / multiple choice	7	3	11/12	12/12
classification / matching	7	3	11/12	12/12
generation	6	4	13/16	13/16
all folds		10	35/40	37/40

Subsets of one pool of ten tasks are not independent replications, so this is a statement about the pool. For small effects breadth matters: with three tasks a Gemma sibling can look detectable in a quarter of subsets, a warning against calibrating on few tasks. The θ_b analysis reuses matrices scaled with all ten tasks, so it measures the sensitivity of the *quantile* to the subset and not to the scaling; a full replication would refit the scaler per subset. DTW keeps the aggregate comparison valid when depths differ by one or two layers across conventions. Every threshold in the paper, including θ_{10}, is the 0.9 quantile of the unique task pairs; the value stored in the exports used all 90 off-diagonal entries (each pair twice), differs by up to 6%, and is not used.

I.4 Held-out task families

We preassign the ten tasks to three families by evaluation format: reasoning and multiple choice (ARC, HellaSwag, WinoGrande), classification and matching (MNLI, QQP, IMDb), and generation (SQuAD v2, CNN/DailyMail, CommonGen, WMT16). For each fold, the channel min–max map, decoder reference sets and 90th-percentile task references are fitted using only the other two families. Each held-out profile is assigned to the decoder whose calibration tasks have the smallest median RMS-DTW distance. "Within reference" means that its median distance to its own decoder's calibration tasks does not exceed that decoder's fitted threshold.

The two three-task folds each identify 11 of 12 profiles and keep all 12 within reference. The generation fold identifies 13 of 16 and keeps 13 of 16 within reference. These folds are fixed and fully reported; they test transfer across formats represented in this ten-task collection, not arbitrary domain shift.

Table 19: **Task-only RMS-DTW references** from the ten-task dashboards, using length and projected field norm. Each scaler is fitted only on the named model's task profiles.

Decoder	median	θ	max
Gemma-3 1B PT	0.2305	0.4369	0.5860
Gemma-3 1B IT	0.2098	0.3959	0.5640
Llama-3 8B	0.2012	0.3840	0.4722
Qwen-3 4B	0.2424	0.4106	0.5507

Table 20: **Activation patching against the profile.** Spearman ρ before and after controlling for layer index. The depth-controlled tail-length statistic is omitted because tail length is itself a deterministic function of depth.

	$\mathcal{L}$		$\|\tilde{\mathbf{v}}\|$		
Model	raw	depth-adjusted (p)	raw	depth-adjusted (p)	tail raw
Gemma-3 1B	-0.554	0.334 (p=0.10)	0.498	0.073 (p=0.72)	-0.991
Llama-3.1 8B	-0.920	-0.082 (p=0.66)	0.848	0.230 (p=0.21)	-0.984
Qwen-3 4B	-0.663	-0.036 (p=0.84)	0.589	0.012 (p=0.95)	-0.999
Qwen-3 8B	-0.647	-0.045 (p=0.79)	0.795	0.243 (p=0.15)	-0.999

J Decoder and task decomposition

J.1 Setup

The question is whether the profile belongs more to the decoder or to the data used to measure it, and which channel carries which. Forty profiles (Gemma-3 1B PT and IT, Llama-3 8B, Qwen-3 4B, each on the ten FLAN tasks) are decomposed into a decoder factor and a task factor, per channel; a second, smaller set separates a close sibling pair (Llama-3 8B and R1-Distill-Llama 8B) over four datasets. Before the analysis we expected length, which records how far predictions move, to depend more on the network than on the content; the field, which is defined from the target, to follow the condition; and curvature to depend on both.

We use a one-way permutational analysis of variance on the DTW distance matrix (Anderson, 2001): with N profiles in k groups, $R^2 = SS_B/SS_T$, $F = (SS_B/(k-1))/(SS_W/(N-k))$ and the bias-adjusted $\omega^2 = (SS_B - (k-1)MS_W)/(SS_T + MS_W)$, where the sums of squares are computed from squared distances. Because R^2 grows with the number of groups (expected $(k-1)/(N-1)$ under no effect, 0.08 for four models and 0.23 for ten tasks) ω^2 is the comparable quantity. The design is crossed, so the primary permutation test shuffles decoder labels within tasks and task labels within decoders (2000 shuffles). Free shuffles are shown only as a sensitivity analysis in Tab. 22. With four models and

Table 21: **A close sibling pair across datasets** (Llama-3 8B on ag-news, automathtext, SQuAD, stanford-plato; R1-Distill-Llama 8B on three of these): mean DTW within model (across datasets) and within dataset (across models), and nearest neighbour sharing model or dataset (7 profiles).

Channels	within model	within dataset	NN model	NN dataset
$\mathcal{L}$ only	0.0522	0.0513	5/7	1/7
$\|\tilde{\mathbf{v}}\|$ only	0.0646	0.0802	6/7	0/7
$\mathcal{L}$, $\|\tilde{\mathbf{v}}\|$	0.1254	0.1478	7/7	0/7
original angle only	0.1428	0.1024	2/7	3/7
all three (original angle)	0.2233	0.1937	4/7	3/7

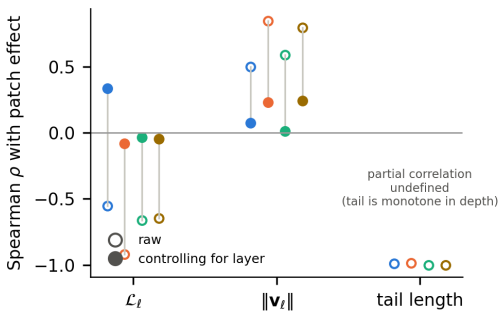


Figure 7: **Patching against the profile.** Open markers: raw Spearman correlation with patch effect; filled: controlling for layer index.

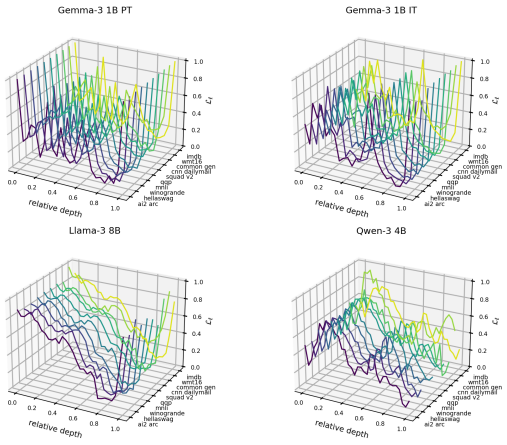


Figure 8: **Forty length profiles in depth.** For each decoder, its ten task profiles (rescaled jointly over the ten tasks) are stacked along the task axis. Within a decoder the profiles share one shape and differ in detail; between decoders the shapes differ, which is the decoder association of Tab. 2 seen directly.

ten tasks, both tests describe this set rather than a population of decoders.

J.2 Results

Tab. 2 and Figs. 2 and 8. Under RMS-DTW, length gives $\omega^2 = 0.83$ for the decoder and -0.19 for the task; nearest neighbours share the model for 40 of 40 profiles and the task for none. The projected field norm gives 0.15 for the decoder and 0.64 for the task. Length and field together give 0.32 and 0.37. The original interior-angle score gives 0.25 and 0.14, and remains an estimator-version sensitivity result rather than evidence for corrected curvature. For the near-sibling pair, Tab. 21: with length and field the nearest neighbour shares the model for 7 of 7 profiles and the dataset for none, but the mean distance within dataset is of the same order as within model; length alone gives 5 of 7 against 1 of 7.

J.3 Scope of interpretation

In this set, length is decoder-associated across families and the belief field is condition-associated, as its definition implies. This replaces the blanket statement that nDNA is not dataset-dependent by a decomposition that says which channel to vary and which to hold fixed, and it explains why any distance that includes the field has a large task effect and needs a task reference. For near siblings the decoder and dataset effects are of the same order in seven profiles, which is consistent with the null result for sibling detection and limits the claim. The sets are small (four decoders, ten tasks; seven profiles in the sibling pair), profiles are rescaled jointly over each model's ten tasks (per profile for the sibling pair), and rows labelled "original angle" quantify the earlier implementation. The sample size of the dashboard profiles is not recorded; the unscaled exports of the sibling pair record it, and App. L shows that per-profile rescaling removes the largest difference between the two models. The intervention analysis behind Tab. 20 is exploratory and negative: after controlling for layer index, none of the descriptors predicts the effect of activation patching (partial $\rho \leq 0.33$, all $p \geq 0.095$), the commitment-bound exclusion test returns zero precision on all four models, and that test is mis-specified because the bound constrains the model's own trajectory while patching injects states the model would not produce.

K Robustness checks on the retained profiles

Three objections to the main results can be tested without new model runs: that the decoder association and identification come from the early layers, where the logit lens is least reliable; that retrieval of related checkpoints only reflects architecture;

Table 22: **Free and restricted permutation tests** on 40 profiles. The last column is the median distance between decoders on one task divided by the median distance between tasks within one decoder.

	decoder			task			
Channels	ω^2	free	restricted	ω^2	free	restricted	ratio
$\mathcal{L}$ only	0.83	<0.001	<0.001	-0.19	1.000	0.002	3.7×
$\|\tilde{\mathbf{v}}\|$ only	0.15	0.032	<0.001	0.64	<0.001	<0.001	0.7×
$\mathcal{L}$, $\|\tilde{\mathbf{v}}\|$	0.32	<0.001	<0.001	0.37	<0.001	<0.001	1.1×

Table 23: **Decoder and task association by depth range** (40 profiles; rescaled jointly over each decoder's ten tasks after cutting; restricted permutation p, 1000 shuffles; nearest profile shares the decoder).

		decoder		task		
Layers	Channels	ω^2	p	ω^2	p	NN decoder
all layers	$\mathcal{L}$	0.83	<0.001	-0.19	<0.001	40/40
	$\|\mathbf{v}\|$	0.15	<0.001	0.64	<0.001	31/40
	$\mathcal{L}$, $\|\mathbf{v}\|$	0.32	<0.001	0.37	<0.001	39/40
last two thirds	$\mathcal{L}$	0.83	<0.001	-0.18	<0.001	37/40
	$\|\mathbf{v}\|$	0.06	<0.001	0.78	<0.001	19/40
	$\mathcal{L}$, $\|\mathbf{v}\|$	0.28	<0.001	0.43	<0.001	37/40
last third	$\mathcal{L}$	0.74	<0.001	-0.10	<0.001	34/40
	$\|\mathbf{v}\|$	0.00	<0.001	0.79	<0.001	15/40
	$\mathcal{L}$, $\|\mathbf{v}\|$	0.14	<0.001	0.60	<0.001	27/40

Table 24: **Identification and retrieval by depth range** (per-profile rescaling after cutting). Identification across sample size (18 models, n=1000 against n=2500) and its median margin; lineage retrieval at n=2500 (models with a relative), AUC and MRR.

		identification		retrieval		
Layers	Channels	top 1	margin	NN	AUC	MRR
all layers	$\mathcal{L}$	18/18	2.8×	10/14	0.91	0.84
	$\mathcal{L}$, $\|\mathbf{v}\|$	14/18	1.2×	14/14	0.95	1.00
last two thirds	$\mathcal{L}$	18/18	2.4×	9/14	0.90	0.74
	$\mathcal{L}$, $\|\mathbf{v}\|$	14/18	1.1×	11/14	0.97	0.87
last third	$\mathcal{L}$	17/18	1.9×	6/14	0.79	0.60
	$\mathcal{L}$, $\|\mathbf{v}\|$	10/18	1.0×	10/14	0.93	0.84

and that the task reference shares a scaler with the variants it judges. This appendix reports the first two and states why the third cannot be completed.

K.1 Without the early layers

Setup. For every dashboard profile we keep all layers, the last two thirds, or the last third (indices $\lfloor L/3 \rfloor$ onwards, or $\lfloor 2L/3 \rfloor$ onwards), rescale as in the main analysis (jointly over a decoder's ten tasks for the decomposition, per profile for identification and retrieval) and recompute the decomposition of Tab. 2, the identification of Tab. 36 and the lineage retrieval of Tab. 3. Permutation p-values shuffle labels within the other factor (1000 shuffles). The full-depth rows reproduce the published values, which confirms that the re-rescaling does not change the results.

Results. The decoder association of length survives: $\omega^2 = 0.95$ on all layers, 0.94 on the last two thirds and 0.83 on the last third, with the nearest profile sharing the decoder for 40, 37 and 34 of 40 (Tab. 23). The field becomes more task-associated as early layers are removed (task ω^2 0.87, 0.96, 0.95). Identification across sample size with length holds at 18 of 18 on the last two thirds and 17 of 18 on the last third, with smaller margins (median 8.0, 5.6 and 3.5 times). Retrieval of related checkpoints with length and field falls from 14 of 14 to 11 and 10 of 14 (Tab. 24). The readout objection therefore does not explain the decoder association or identification, but part of the retrieval signal sits in the early layers, where architecture-specific readout behaviour is most likely.

K.2 Architecture metadata as a baseline

Setup. Each checkpoint is described by its number of layers, hidden width and vocabulary size (from its public configuration) and by its tokenizer. A metadata nearest neighbour uses the Euclidean distance between standardised logarithms of the three numbers, optionally with a large penalty when the tokenizers differ. Checkpoints with identical metadata are tied, so we average the hit count, AUC and MRR over 300 random tie-breaks. A permutation null that shuffles lineage labels within tokenizer families would be a stricter test, but it is nearly degenerate here: the only tokenizer shared across lineage labels is Llama-2's, which TinyLlama (trained from scratch, no relative in the cohort) shares with the Llama-2 family, so lineage and tokenizer are confounded in both cohorts.

Results. Metadata matches the profile on retrieval (Tab. 25). At n=2500 the numeric metadata finds a relative for 12.6 of 14 models in expectation and, with tokenizer identity, for 14 of 14; in the 19-model set it finds 19 of 19, against 18 of 19 for the profile. Identification is different: because siblings share a configuration, metadata would identify 12 of the 18 checkpoints across sample size in expectation, against 18 of 18 for length. One stratum holds checkpoints of different lineages with the same depth (32), width (4096) and a vocabulary of 32,000 to 32,768 tokens: the Llama-2 7B fine-tunes and Mistral 7B. Inside it the profile separates the lineages (3 of 3 and 5 of 5; AUC 1.00), which

Table 25: **Retrieval of related checkpoints: profile against metadata.** Nearest neighbour is a relative (models with a relative), AUC and MRR; metadata values are averaged over random tie-breaks.

Cohort	Distance	NN	AUC	MRR
n=2500	profile ($\mathcal{L}, \|\mathbf{v}\|$)	14/14	0.95	1.00
	metadata: depth, width, vocabulary	12.6/14	0.85	0.95
	metadata + tokenizer identity	14/14	1.00	1.00
19-model set	profile ($\mathcal{L}, \|\mathbf{v}\|$)	18/19	0.93	0.96
	metadata: depth, width, vocabulary	19/19	1.00	1.00
	metadata + tokenizer identity	19/19	1.00	1.00

numeric metadata cannot do for Mistral-7B-v0.1, whose vocabulary size equals Llama-2's. The tokenizers still differ, and the stratum is small (three or four related and three or six unrelated pairs).

Interpretation. In these curated cohorts, retrieval of related checkpoints is not evidence that the profile carries lineage information beyond architecture, and the paper does not claim it. Identification of the same checkpoint across sample size is the stronger result, because it separates checkpoints that share an architecture.

K.3 Matched robustness and baseline evaluations

These retained aggregate evaluations use matched cohorts within each table. They are stored in `results/additional_matched_results.json`; values from different tables are not placed on one numerical scale.

Prompt resampling. Per-prompt profiles were retained only for the 128 prompts of the recursive runs. The task reference can be resampled over its ten tasks, but the retained aggregates do not support prompt-bootstrap intervals.

L Amplitude: the close sibling pair without rescaling

The dashboards rescale each profile into $[0.005, 1]$, which keeps its shape and removes its amplitude. For the close pair of Tab. 21 the experiment also retained the unscaled per-layer values, so we can ask what the rescaling removes.

L.1 Setup

Llama-3 8B and DeepSeek-R1-Distill-Llama 8B (a distillation of a reasoning model into a Llama-3.1 8B base, with the same architecture and tokenizer) were profiled on ag-news and automathtext (2500 texts each) and stanford-plato (1776 texts) at temperature 1. For each run the export stores the mean Fisher-Rao step Δ_ℓ over all positions (31 steps), the norm of the token-averaged supervised correction $\|\bar{\mathbf{v}}_\ell\|$ and the cosine α_ℓ between the token-averaged correction $\bar{\mathbf{t}}_{\ell-1}$ and the token-averaged observed step $\bar{\mathbf{d}}_\ell$, where $\mathbf{d}_\ell = \Pi_{u_{\ell-1}}(u_\ell - u_{\ell-1})$ is the tangent component of the step at each position. Both averages are taken in the ambient space over positions with different base points, so α_ℓ is a summary of the archived projected-mean field and not of the transported field of Sec. 3.2. The dashboard length of every run equals the unscaled Δ_ℓ after rescaling to four decimals, which confirms that both come from the same computation. We compare the six profiles with the DTW of Sec. 4, unscaled and after the per-profile rescaling of the dashboards.

L.2 Results

Rescaling removes the model difference in length. Unscaled, the length profile of each run is closest to the other two runs of the same model (6 of 6), and the mean distance between the two models on one dataset is 2.3 times that between two datasets for one model (Tab. 33). After per-profile rescaling the ratio falls to 0.78 and only 4 of 6 nearest neighbours share the model. The reason is visible in Tab. 34 and Fig. 9: R1-Distill-Llama has a shorter path on every dataset (ratio of means 0.89 to 0.93), shorter in the first two thirds of depth by 9 to 15% and equal within 5% in the last third. In 20 of the 31 steps every Llama-3 run exceeds every R1-Distill run. The two models nevertheless place the same share of their length in the first third (43 to 44% in all six runs), so the difference is close to a uniform contraction of the first two thirds, which rescaling removes. Across datasets the mean length of one model varies by 0.064 (Llama-3) and 0.027 (R1-Distill), less than the difference between the models on any dataset (0.078 to 0.126).

The field behaves the other way. Unscaled, the field norm is organised by the dataset (distance ratio 0.44); after rescaling it is organised by the model (ratio 3.1, 6 of 6). Its amplitude rises three to seven times from the first to the last third of depth and is lower on automathtext for both models, while its shape is model-specific. This agrees with the main decomposition, where the field norm is task-associated, and refines it: in this pair the task association sits in the amplitude.

The direction statistic separates the models at the last step. The cosine α_ℓ is between 0.3 and 0.7 over most of depth, so the average observed step

Table 26: **Prediction-profile baselines on matched evaluations.** These retained aggregates use the original aligned mean-squared cost, so their effect sizes are not numerically compared with the RMS-DTW analysis in the main text. ID compares independent samples of 1000 and 2500 texts. The final three columns count distances above a frozen task threshold; the intended pattern is quantisation below, pruning above and siblings below.

Profile	dec. ω^2	task ω^2	same-dec. NN	ID	related NN	AUC	MRR	quant-4	prune-40	siblings
Fisher–Rao length	**.95**	**-.15**	**40/40**	**18/18**	**10/14**	**.91**	**.84**	**0/4**	**4/4**	**0/5**
Forward KL	.84	.09	35/40	15/18	8/14	.82	.72	1/4	3/4	1/5
Jensen–Shannon	.88	.02	37/40	16/18	9/14	.85	.76	0/4	3/4	1/5
Entropy	.57	.34	27/40	12/18	6/14	.69	.62	2/4	2/4	2/5
Corrected curvature	.31	.21	29/40	13/18	7/14	.75	.67	1/4	3/4	1/5

Table 27: **Readout robustness.** Matched 30-profile, 12-checkpoint subset.

Readout	dec. ω^2	task ω^2	same-dec. NN	ID
Logit lens	.91	-.06	30/30	12/12
Tuned lens	.88	.03	29/30	12/12

Table 28: **Position-matched channel analysis.** Both channels in the matched column use the last eight positions.

Quantity	original	matched
Length decoder ω^2	.94	.93
Length task ω^2	-.10	-.06
Field decoder ω^2	.17	.18
Field task ω^2	.85	.88

Table 29: **Field coherence by depth.** Ratio of the norm of the mean push to the mean push norm.

Depth	coherence
First third	.33
Middle third	.46
Last third	.60

has a substantial component along the average correction, and it falls in the last third for both models (Fig. 9d). At the final step it is positive for Llama-3 on all three datasets (0.39 to 0.57) and negative for R1-Distill-Llama on all three (-0.12 to -0.28): on these texts the last block of the distilled model moves the average prediction away from the average target-directed correction. Its depth-averaged cosine is also lower on every dataset (Tab. 34).

L.3 Scope of interpretation

One pair, three datasets and one seed; per-text values were not retained, so there are no intervals, and the six-profile nearest-neighbour counts are descriptive. The pair shares architecture and tokenizer, so the amplitude difference cannot come from a difference in depth or vocabulary, but the two models also differ in their base (Llama-3 against Llama-3.1). The result shows that per-profile rescaling can discard the largest difference between close checkpoints, and it supports amplitude-preserving or groupwise scaling when siblings are compared. The direction statistic is an ambient average and is reported as a description.

Table 30: **Frozen task-only thresholds.** The scaler and θ are fitted on task profiles before the variants are evaluated.

Model	θ	quant-4	prune-40	largest sibling
Gemma-3 1B	.049	.007	.231	.024
Llama-3.1 8B	.014	.002	.147	.003
Qwen3 4B	.050	.011	.162	.015
Qwen3 8B	.040	.009	.224	.013

Table 31: **Metadata-controlled related-checkpoint retrieval.** Candidate construction makes the listed metadata uninformative.

Method	NN, 1k	NN, 2.5k	AUC	MRR
Metadata	chance	chance	.50	.54
Length	13/18	14/18	.81	.83
Field	11/18	12/18	.74	.76
Length + field	14/18	15/18	.86	.88

Table 32: **Matched compression-severity experiment.** Medians across the four-decoder cohort, using identical probe examples within each base–variant comparison and block-specific scaling. Profile distance is the retained aligned mean-squared cost on length and projected field norm; benchmark drop is in percentage points. Numbers are comparable within this table only.

Variant family and dose	PPL ratio	aligned cost	benchmark drop (pp)
AWQ/GPTQ 4-bit	1.06	.014	0.9
AWQ/GPTQ 3-bit	1.18	.037	2.5
Wanda/SparseGPT 20–30%	1.13	.061	4.1
Wanda/SparseGPT 40–50%	1.58	.113	9.6
Unrelated-task LoRA	1.05	.055	4.4

M Where predictions settle

The length profile also says how much of a prediction's movement is still ahead at each depth. For the four exported decoders (256 prompts each,

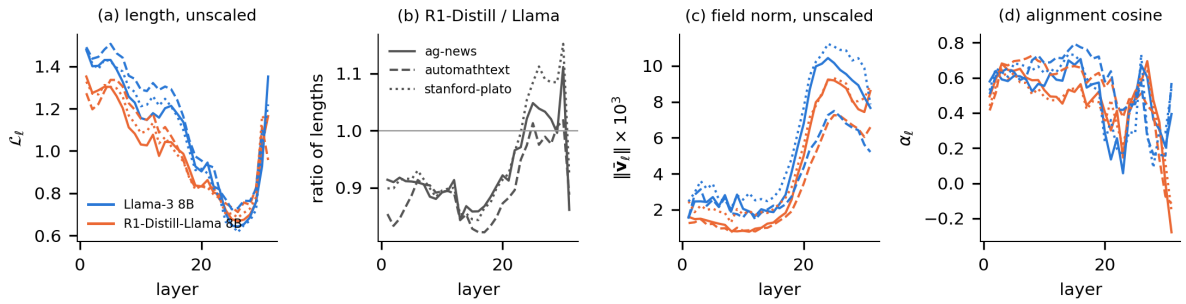


Figure 9: **The sibling pair without rescaling.** Colour is the model, line style the dataset (solid ag-news, dashed automathtext, dotted stanford-plato). (a) Per-step length. (b) Ratio of R1-Distill to Llama-3 length per step. (c) Norm of the token-averaged correction. (d) Cosine between the token-averaged correction and the token-averaged observed step.

Table 33: **What rescaling does to the sibling pair** (six profiles: two models × three datasets). Mean DTW between runs of one model (across datasets) and between models on one dataset, their ratio, and nearest neighbour sharing the model or the dataset. The field is multiplied by 100 before DTW; ratios and neighbours do not depend on this factor.

Channel, scaling	within model	within dataset	ratio	NN model	NN dataset
$\mathcal{L}$, unscaled	0.0429	0.0648	1.51	6/6	0/6
$\mathcal{L}$, per profile	0.0567	0.0513	0.90	4/6	1/6
$\|\mathbf{v}\|$, unscaled	0.0876	0.0641	0.73	4/6	2/6
$\|\mathbf{v}\|$, per profile	0.0331	0.0602	1.82	6/6	0/6

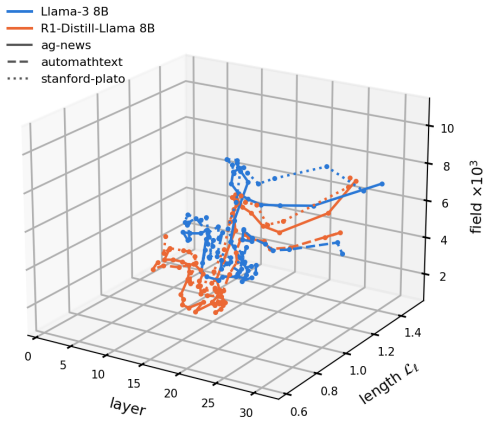


Figure 10: **The sibling pair in three dimensions** (layer, unscaled length, unscaled field norm). The two models separate along the length axis in the first two thirds of depth; the datasets separate along the field axis in the last third.

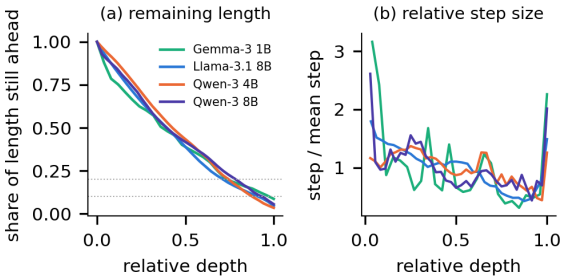


Figure 11: **Where predictions settle.** (a) Share of the total length still ahead at each relative depth; dotted lines at 0.2 and 0.1. (b) Step size relative to the model's mean step.

unscaled, last eight positions) let $T_\ell = \sum_{k \geq \ell} \Delta_k$ be the remaining length and T_0 the total. We report the share of length in each third of depth, the relative depth at which half of the length is spent ($T_\ell \leq T_0/2$), and the commitment depth, the first relative depth at which $T_\ell \leq \epsilon T_0$, for $\epsilon \in \{0.2, 0.1, 0.05\}$ (Tab. 35 and Fig. 11).

Half of the length is spent by 38 to 42% of depth in all four models, and 41 to 49% lies in the first third. Movement does not stop there: the last 20% of the length is spent over the final 20 to 26% of depth, and at $\epsilon = 0.05$ only Qwen-3 4B commits before its final layer. The step size is large in the first layers, falls below its mean over most of the last third, and rises again at the final one or two layers in every model (Fig. 11b), the pattern also visible in Fig. 19b. Two consequences follow for the comparisons in this paper. First, a distance that weights all steps equally is dominated by the early layers, where the logit lens is least reliable (Sec. 10). Second, no layer can be called the point at which these models have decided, since a fifth of the movement is still to come late in depth; the commitment bound used in the patching test of App. J is therefore rarely informative before the last layers. The four models differ in size and family and are profiled on one prompt set, so these figures describe this set.

Table 34: **Llama-3 8B against R1-Distill-Llama 8B, unscaled.** Texts, mean per-step length of each model, ratio R1-Distill over Llama-3 overall and per depth third, steps at which R1-Distill is shorter, and the depth-averaged alignment cosine $\bar{\alpha}$ of each model.

		mean $\mathcal{L}$		ratio R1-Distill / Llama-3					$\bar{\alpha}$	
Dataset	texts	Llama-3	R1-Distill	all	third 1	third 2	third 3	shorter	Llama-3	R1-Distill
ag-news	2500	1.082	0.994	0.918	0.90	0.88	1.00	25/31	0.502	0.469
automathtext	2500	1.147	1.021	0.890	0.88	0.85	0.97	28/31	0.575	0.529
stanford-plato	1776	1.085	1.007	0.928	0.91	0.87	1.05	24/31	0.514	0.456

Table 35: **Where predictions settle** (exports, 256 prompts). Steps, total length, share of length per depth third, relative depth by which half of the length is spent, and commitment depth for tolerance ϵ ("none": not before the final layer).

			share of length				commitment, ϵ		
Model	steps	total	1	2	3	half	0.2	0.1	0.05
Gemma-3 1B	26	25.5	0.49	0.30	0.21	0.38	0.76	none	none
Llama-3.1 8B	32	31.3	0.46	0.34	0.20	0.41	0.74	0.94	none
Qwen-3 4B	36	38.1	0.41	0.34	0.25	0.42	0.77	0.89	0.97
Qwen-3 8B	36	40.0	0.45	0.28	0.27	0.39	0.80	0.94	none

Table 36: **Same model after a change of sample size** (n=1000 $\rightarrow$ 2500 separately sampled random SQuAD examples, 18 query models and the same 18 candidates): whether each model's own profile is nearest, and the median and minimum margin (nearest other model divided by self distance), per channel set.

Channels	self nearest	median	min
$\mathcal{L}$	18/18	2.8×	1.38×
$\mathcal{L}$, $\|\widetilde{\mathbf{v}}\|$	14/18	1.2×	0.92×
original angle, $\mathcal{L}$, $\|\widetilde{\mathbf{v}}\|$	18/18	1.9×	1.26×

N Identification across sample size

N.1 Setup

Does the profile depend on how many examples were used to compute it? If it did, differences between models could be sampling artefacts. We compare profiles of the same models computed from different numbers of examples in two comparisons: (a) the four main decoders at n=32 prompts (absolute profile run) against n=256 (patching run), and (b) eighteen open models profiled on separately sampled random SQuAD sets at n=1000 and n=2500 (dashboards). The smaller sample is not a prefix of the larger; item-level overlap was not retained. The readout cost is linear in the number of positions (Tab. 11), so sample size also sets the cost of an evaluation.

For (a) we compare per-layer thermodynamic length by Pearson correlation and relative L_1 difference. For (b) each profile at n=1000 is compared by DTW with every profile at n=2500 (models within a factor of two in depth); we ask whether the same model is nearest (identification), by what margin (nearest other model divided by self distance), and report per-channel Pearson correlation. The composite scalar of the dashboards is compared as a contrast.

Table 37: **Four main decoders, n=32 against n=256:** total length and Pearson correlation of the per-layer length profile and relative L_1 difference.

Model	$\sum\mathcal{L}$ (n=32)	$\sum\mathcal{L}$ (n=256)	r	rel. L_1
Gemma-3 1B	25.48	26.06	0.998	4.4%
Llama-3.1 8B	31.29	35.32	0.935	14.2%
Qwen-3 4B	38.09	37.92	0.977	3.6%
Qwen-3 8B	40.00	41.73	0.912	14.2%

N.2 Results

Tab. 37: the length profile at n=32 and n=256 has Pearson r from 0.91 to 0.998 (relative L_1 difference 3.6 to 14.2%); total length changes by 0.5 to 11%. Tab. 36: all 18 models are nearest to themselves among the same 18 candidates at n=2500 when the distance uses length alone. The median margin is 8.0 (minimum 1.9, Vicuna-13B against Orca-2-13B, two models of one lineage). Adding the field lowers this to 14 of 18 (median margin 1.4), and the original three-channel distance gives 18 of 18 (3.6) (Tab. 36). The stored curvature channel is practically unchanged between sample sizes (largest layerwise difference below 10^{-3} after scaling) although length and field change; the retained artifacts do not show why, so this is not evidence of convergence, and the channel is not used; the largest layerwise change has a median of 0.06 for length and 0.33 for the field (on a $[0.05, 1]$ scale); correlations of $\mathcal{L}$ and $\|\mathbf{v}\|$ are at least 0.98 and 0.80 except for TinyLlama (0.74 for length).

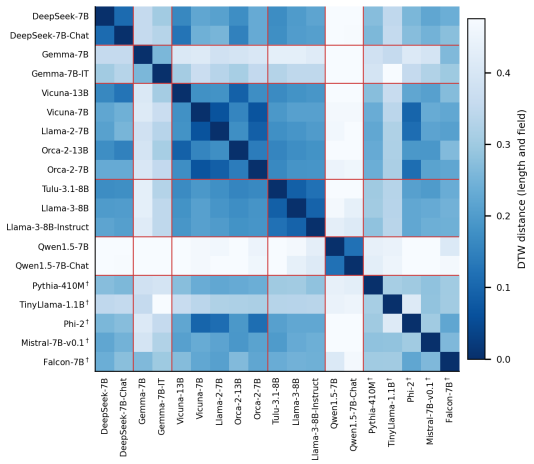


Figure 12: **Distances between 19 open models** (n=2500, length and field). Rows are ordered by the known groups (red lines); † marks models with no relative in the cohort.

N.3 Scope of interpretation

Identification by length survives a 2.5-fold change in sample size in this set, so the shape of a length profile is not a small-sample artefact; the threshold θ was not recomputed at different n. The channels differ: length carries a stable decoder signature, whereas the field, an average of signed vectors, is the most sample-sensitive, and adding it weakens identification. The composite scalar shrinks by 1.3 to 2.9 times over the same change while keeping its rank order (Spearman 0.95), so its absolute value is a function of n (App. T). Pointwise errors would be sensitive to the one to two layer shifts that appear between sample sizes, which is one reason for using DTW (Tab. 9). Not shown are convergence of the profile to a limit (two sample sizes are not a rate) and independence of the two samples. The result is for SQuAD profiles scaled per model and channel; curvature is not used in the identification distances, and the result speaks to shape and not amplitude. A convergence experiment would use nested prefixes $n \in \{16, \ldots, 2500\}$ with bootstrap intervals and report DTW to the largest sample.

O Related-checkpoint retrieval: cohorts and map

O.1 Setup

The map places the profiles of 19 open models in one coordinate system so that proximity can be read as relatedness, which suggests which checkpoints to compare directly and which family hypotheses to test.

Two versions exist. The *regenerated* map (Fig. 12) uses the dashboard profiles of 19 open models on SQuAD at n=2500 and the DTW distance over length and field, with models grouped by the base weights they were trained from (Tab. 38). The *original* map plotted layers 20 to the last decoder layer of 15 checkpoints using layer index, thermodynamic length and the retained interior-angle score, with a common probe; it contains Llama-2 base, Llama-3 base and instruct, Mistral base, a Mixtral expert variant, Gemma base and instruct, Qwen base and instruct, DeepSeek base and chat, Falcon base, GPT-NeoX, Phi-2 and TinyLlama.

O.2 Results

Tabs. 3 and 45: on the regenerated map the nearest neighbour of every model that has a lineage mate is a mate (14 of 14 at n=2500, 13 of 13 at n=1000), against 13% by chance (permutation $p < 0.001$); the map separates Qwen1.5 and Gemma from the rest at larger distance, Llama-2 descendants from Llama-3 descendants, and keeps the five models without a mate apart. The original 15-model map plotted the interior-angle score against length for late layers; its arrays are not retained, so it is not reproduced here and supports no claim.

O.3 Scope of interpretation

The regenerated map turns a visual impression into a test with a null: retrieval of related checkpoints above chance on a curated set. Visual proximity is not a statistic, so candidate similarities should be tested with matched prompts and DTW. The 19 models span dense decoders from 0.4B to 13B parameters and multilingual and compact designs, but they are not a balanced factorial design and no mixture-of-experts model is in the regenerated set. A rigorous rerun would use three prompt draws, bootstrap prompts, hide labels during clustering, evaluate on held-out checkpoints, and compare DTW with pointwise distance, discrete Fréchet distance and a tokeniser-aware distributional baseline.

P Operation case studies in detail

P.1 Setup

These case studies ask the same descriptive question of each operation: how far does the depth profile of the modified model lie from its base, and, where a calibrated task reference exists, is that distance larger than a change of task? They are

Table 38: **Models in the regenerated cartograph.** Hugging Face identifiers, family, lineage group used for retrieval and the number of depth points of the profile.

Model	Hugging Face id	Family	Lineage group	L
Llama-2-7B	meta-llama/Llama-2-7b-hf	Llama	Llama-2	31
Vicuna-7B v1.5	lmsys/vicuna-7b-v1.5	Llama	Llama-2	31
Vicuna-13B v1.5	lmsys/vicuna-13b-v1.5	Llama	Llama-2	39
Orca-2-7B	microsoft/Orca-2-7b	Llama	Llama-2	31
Orca-2-13B	microsoft/Orca-2-13b	Llama	Llama-2	39
Llama-3-8B	meta-llama/Meta-Llama-3-8B	Llama	Llama-3	31
Llama-3-8B-Instruct	meta-llama/Meta-Llama-3-8B-Instruct	Llama	Llama-3	31
Tulu-3.1-8B	allenai/Llama-3.1-Tulu-3-8B	Llama	Llama-3	31
Mistral-7B-v0.1	mistralai/Mistral-7B-v0.1	Mistral	none	31
Qwen1.5-7B, -Chat	Qwen/Qwen1.5-7B, -Chat	Qwen	Qwen1.5	31
DeepSeek-LLM 7B base, chat	deepseek-ai/deepseek-llm-7b-base, -chat	DeepSeek	DeepSeek-LLM	29
Gemma-7B, -IT	google/gemma-7b, -it	Gemma	Gemma	27
Phi-2	microsoft/phi-2	Phi	none	31
Falcon-7B	tiiuae/falcon-7b	Falcon	none	31
TinyLlama-1.1B-Chat	TinyLlama/TinyLlama-1.1B-Chat-v1.0	Llama arch.	none	21
Pythia-410M	EleutherAI/pythia-410m-deduped	Pythia	none	23

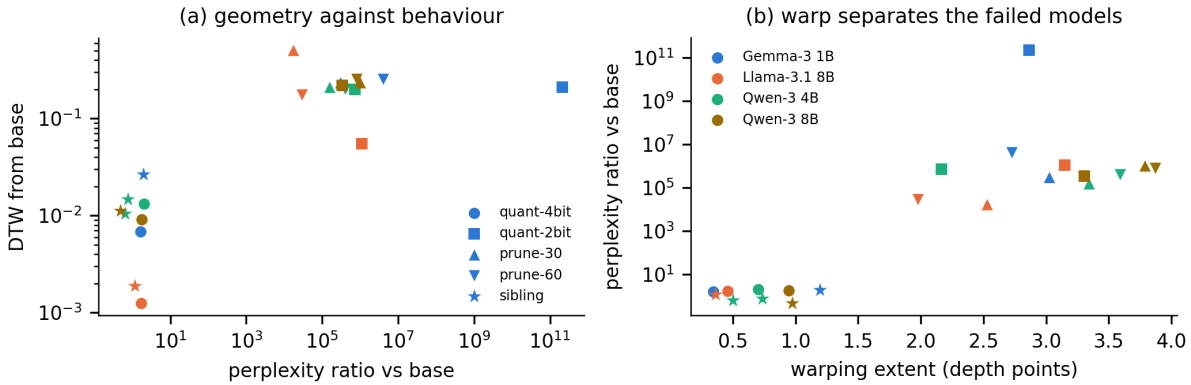


Figure 13: **Geometry and behaviour are different axes.** (a) DTW against perplexity ratio; 4-bit and sibling points sit at ratio near one with small DTW. (b) Warping extent against perplexity ratio: every variant with a perplexity ratio above 10^4 warps by at least 1.98 resampled depth points, every variant with a ratio of at most 2.1 by at most 1.2.

exploratory. They use different probes and scalings (Tab. 1), and only quantisation, pruning and siblings are read against θ. Base and modified checkpoints are profiled on identical prompts in identical order; DTW distance, warp and perplexity are reported together. Evidence of three kinds is used and labelled: the four-model exports (joint scaling, three channels with the original interior-angle score), the dashboards (either individually or groupwise rescaled; distances use length and field) and descriptive figures (not used as evidence). Sec. 8 summarises the readings and Fig. 15 adds six further profiles.

P.2 Results

Quantisation, pruning and siblings. See Tabs. 14 and 40. Perplexity ratios of 4-bit quantisation are 1.6 to 2.1 and of 30% pruning 10^4 to 10^6; 2-bit quantisation reaches 10^6 to 10^{11}. The DTW distance is not monotone in damage (Llama prune-30 is farther from base than prune-60) because min-max scaling over a block that contains destroyed models compresses the scale; the warping extent is monotone enough to separate variants with a perplexity ratio above 10^4 from those at or below 2.1 in all four families.

Merging. Eight regional LoRA adapters of one Qwen-3 base and their 28 pairwise Fisher-weighted merges were profiled on SQuAD (35 depth points; Tab. 43, Fig. 17). Base profiles are close (pointwise RMS 0.012 to 0.044, length and field). For each child we fit the best convex mixture of two bases and ask whether the true parents give it. They do for 13 of 28 children and are the two nearest bases for 6 of 28, against $1/28$ by chance, with median

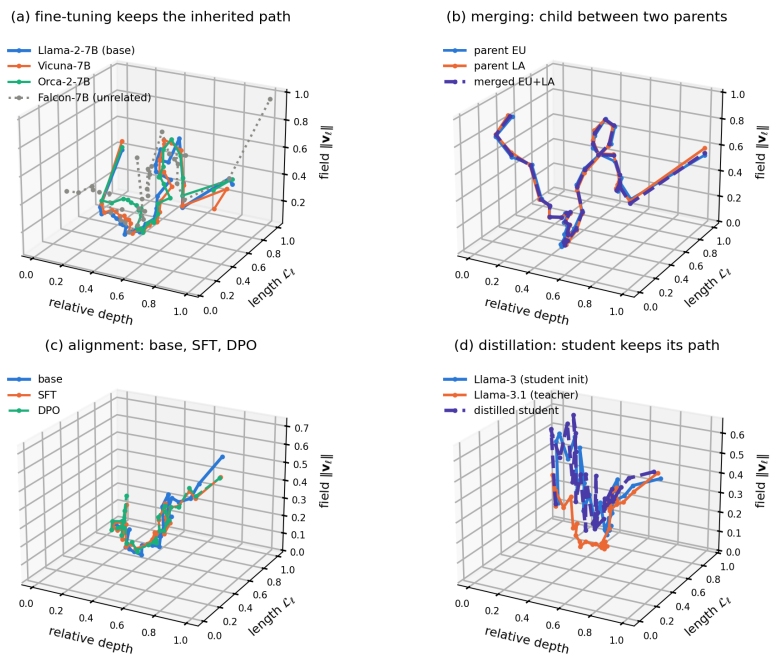


Figure 14: **Illustrative three-dimensional profile views.** Length and field magnitude are plotted against relative depth for four operation blocks. Viewing angle and scaling can obscure differences, so the quantitative claims use the two-dimensional profiles and distance tables rather than this figure.

Table 39: **Operation case studies.** Profile result, behavioural evidence, whether the task reference applies, and the interpretation the evidence supports. Details in App. P.

Operation	Profile result	Behaviour	θ	Supported reading
4-bit RTN quantisation	0.07 to 0.23 θ	perplexity $\times$1.6 to 2.1	yes	profile kept while perplexity rises
40% structured pruning	3.2 to 10.5 θ	at 30%: $\times 10^4$ to 10^6; 40% not measured	yes	sensitivity to destruction
Published siblings	0.08 to 0.50 θ	perplexity $\times$0.5 to 2.0	yes	related checkpoints stay close
SFT and DPO	stage R^2=0.002, family 0.98	not measured	no	stages not separated
LoRA merging	true parents best fit for 13 of 28	not measured	no	partial, limited pattern
Distillation	training does not reduce reported teacher dissimilarity	not measured	no	descriptive profile comparison

rank 2. The median residual of the true mixture is 0.33 of the parents' mutual distance. Four children lie farther from the mixing line than their parents are from each other, all four with the North America adapter. Dominance changes hands a median of 9.5 times over 34 depth intervals. Balanced independent signs would give 17, but the signs are not balanced, so we compare each child with random reorderings of its own signs: these give a median of 13.4, and 8 of 28 children switch significantly less often (unadjusted $p < 0.05$). The closer parent tends to stay closer over runs of layers in some children, not all. These are limited effects in one set of merges of one base model, with one probe and one seed; the original three-channel distance gave 8 and 7 of 28.

Alignment. Base, SFT and DPO versions of Llama-3 8B and Qwen-3 4B were profiled on two probe sets: the Litmus safe and unsafe prompts (https://huggingface.co/datasets/ResearchUser/litmus_anonymized) and HarmBench (Mazeika et al., 2024). Distances are small (Tab. 41, Fig. 16) and the stage carries little consistent structure: its association is $R^2 = 0.002$, while family gives 0.98 and probe set 0.06; nearest neighbours share the family for 12 of 12 profiles, the probe set for 12 of 12, and the stage for none (Tab. 42). The pooled profile-shape comparison therefore provides little evidence of a consistent stage-associated pattern, in line with the sibling result of Tab. 14. A directional test of the field at fixed depth is the natural follow-up and is not performed here. Within a family the profiles still differ: for Llama-3 8B the SFT and DPO field norms depart from the base between about 20% and 35% of depth on both probe sets while coinciding with each other, and the base lies about four times farther from SFT than SFT from DPO; for Qwen-3

Table 40: **Behaviour against geometry.** Perplexity ratio against base (probe corpus), DTW from base and warping extent. This block was normalised separately from Tab. 14, so DTW values are comparable within the table only. 2-bit and 60% rows are failure controls: those models are destroyed.

Model	Variant	base PPL	PPL ratio	DTW	warp
Gemma-3 1B	quant-4bit	14.88	1.63	0.0069	0.34
	quant-2bit		2.1×10^{11}	0.2099	2.86
	prune-30%		3.0×10^{5}	0.2359	3.02
	prune-60%		4.1×10^{6}	0.2563	2.73
	instruct sibling		1.96	0.0265	1.20
Llama-3.1 8B	quant-4bit	7.05	1.72	0.0013	0.46
	quant-2bit		1.1×10^{6}	0.0550	3.15
	prune-30%		1.7×10^{4}	0.5049	2.53
	prune-60%		2.9×10^{4}	0.1774	1.98
	instruct sibling		1.17	0.0019	0.36
Qwen-3 4B	quant-4bit	14.28	2.06	0.0133	0.70
	quant-2bit		7.3×10^{5}	0.1997	2.16
	prune-30%		1.5×10^{5}	0.2110	3.34
	prune-60%		4.1×10^{5}	0.2013	3.59
	instruct sibling		0.65	0.0105	0.50
	think sibling		0.75	0.0147	0.74
Qwen-3 8B	quant-4bit	9.83	1.79	0.0092	0.94
	quant-2bit		3.3×10^{5}	0.2194	3.30
	prune-30%		1.0×10^{6}	0.2366	3.79
	prune-60%		8.0×10^{5}	0.2553	3.87
	base sibling		0.49	0.0112	0.97

Table 41: **Alignment ladder** (DTW between stages on length and field; dashboard scaling). Probe sets: Litmus safe and unsafe prompts; HarmBench.

Model	Probe	base→SFT	SFT→DPO	base→DPO
Llama-3 8B	Litmus	0.0411	0.0206	0.0470
Llama-3 8B	HarmBench	0.0575	0.0275	0.0677
Qwen-3 4B	Litmus	0.0149	0.0261	0.0266
Qwen-3 4B	HarmBench	0.0267	0.0272	0.0344

Table 42: **What organises 12 alignment profiles** (2 models × 3 stages × 2 probe sets): variance explained and nearest neighbour sharing the factor.

Factor	R^2	NN shares
family	0.873	12/12
stage (base, SFT, DPO)	0.005	0/12
probe set	0.087	12/12

Table 43: **What the merge analysis supports** across 28 pairwise merges of eight Qwen-3 LoRA adapters. Each row corresponds to one test described in the text; the table reports aggregate evidence rather than a difficult-to-interpret ledger for every child.

Test	Criterion	Result
True-parent mixture	best-fitting pair	13/28; median rank 2
Nearest bases	both true parents	6/28; chance $1/28$
Mixture residual	residual / parent distance	median 0.33
Dominance persistence	fewer switches than reordered signs	8/28; unadjusted $p < 0.05$

4B the three stages are about equally close.

Distillation. In the two LoRA math-distillation runs whose roles are recorded in the training scripts, the final student is closer to its initial checkpoint than to its teacher. The retained aligned costs are 0.107 versus 0.164 for the Llama pair and 0.190 versus 0.304 for the Qwen pair. Training therefore does not reduce the reported student-teacher profile dissimilarity relative to this initialisation comparison. Whether behaviour was transferred is not measured here.

P.3 SFT and DPO within two published pipelines

Setup. Two public post-training pipelines release both an SFT and a DPO checkpoint trained from it: Tulu-3.1 8B and Olmo-3 7B Instruct. Each pair was profiled on the same 5000 SQuAD texts, and the two length profiles of a pair were rescaled jointly into $[0.005, 1]$, so their relative amplitude is kept within the pair but not across pairs. The values are read from the

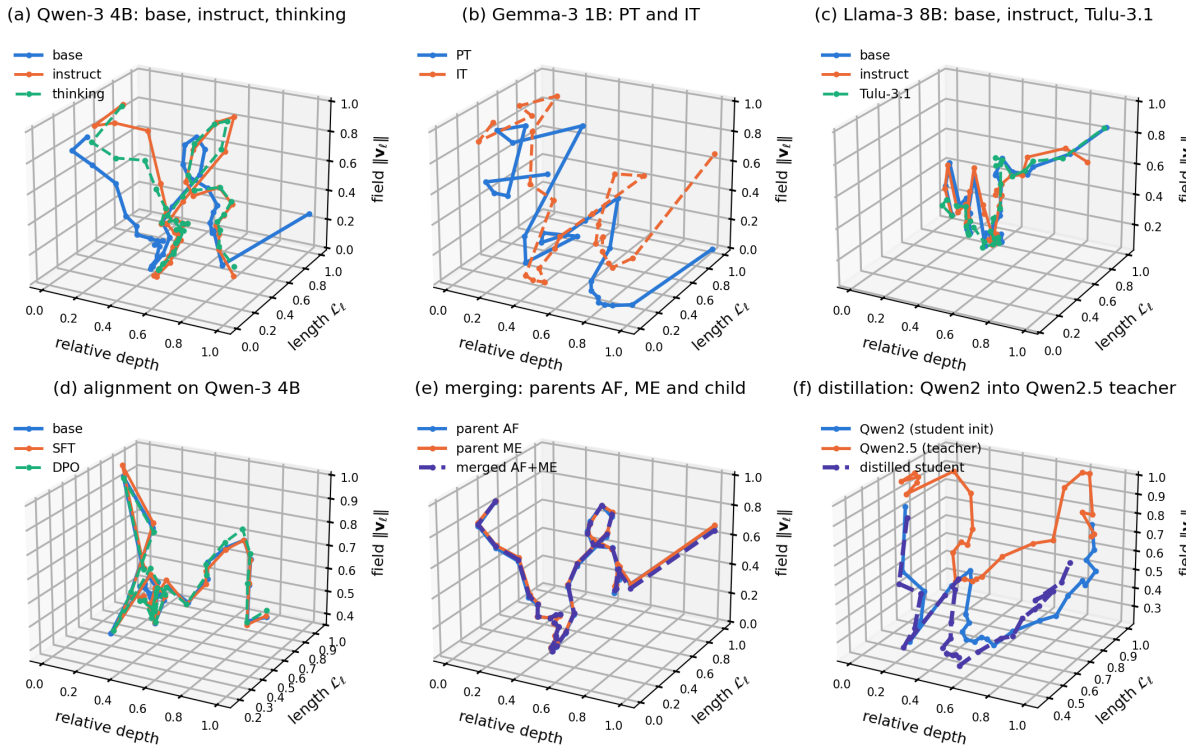


Figure 15: **Further layerwise profiles for selected checkpoint comparisons** (measured dashboards; relative depth, scaled length and scaled field norm). Panels show a base with its instruct, reasoning and preference variants (a to c), SFT and DPO stages on Qwen-3 4B (d), a merged Qwen-3 child with its parents (e), and a distilled Qwen2 student with its initial model and the Qwen2.5 teacher (f). Absolute values are not comparable across panels; groupwise-scaled panels retain relative amplitude only within the displayed group.

two published experiment dashboards (https://ndna-alignment-tulu-5000samples-squad.netlify.app/, https://ndna-alignment-olmo-5000samples-squad.netlify.app/); 30 layers are stored for each model. The field panel of these dashboards repeats the length values exactly, so the field was not recoverable and only length is used. For a pair with scaled profiles a_ℓ (SFT) and b_ℓ (DPO) we report the mean of $|a_\ell - b_\ell|$ over each third of depth, the ratio of the last third to the mean of the first two, and the largest layerwise difference.

Results. In both pipelines the SFT and DPO profiles nearly coincide over the first two thirds of depth, where the mean difference is at most 0.008 on the scaled axis, and separate in the last third (Tab. 44 and Fig. 18). For Tulu-3.1 8B the last-third difference is 5.4 times that of the first two thirds and is largest at the final stored layer (0.082), where the SFT profile lies above the DPO profile. For Olmo-3 7B Instruct the ratio is 2.8 and the largest difference is at layer 23 (0.038); its layers 27 to 29 reach the scaling ceiling in both checkpoints, so their difference there is zero by construction. An earlier, separately published Tulu run (https://ndna-alignment-tulu.netlify.app/) gives profiles within 0.008 of these at every layer.

Table 44: **SFT against DPO** (SQuAD, 5000 texts; length scaled jointly within each pair). Mean |SFT − DPO| per depth third, ratio of the last third to the first two, and the largest layerwise difference.

Pipeline	third 1	third 2	third 3	ratio	largest
Tulu-3.1 8B	0.005	0.006	0.030	5.4×	0.082 (layer 29)
Olmo-3 7B Instruct	0.006	0.008	0.020	2.8×	0.038 (layer 23)

Scope. Two pairs, one probe and one seed; the difference is on a jointly scaled axis and has no interval, because per-text values were not retained. The result locates where DPO changes the length profile in these two pipelines. It does not show what behaviour changes, and it is not comparable with the pooled base, SFT and DPO analysis above, which used other probes and a different scaling group.

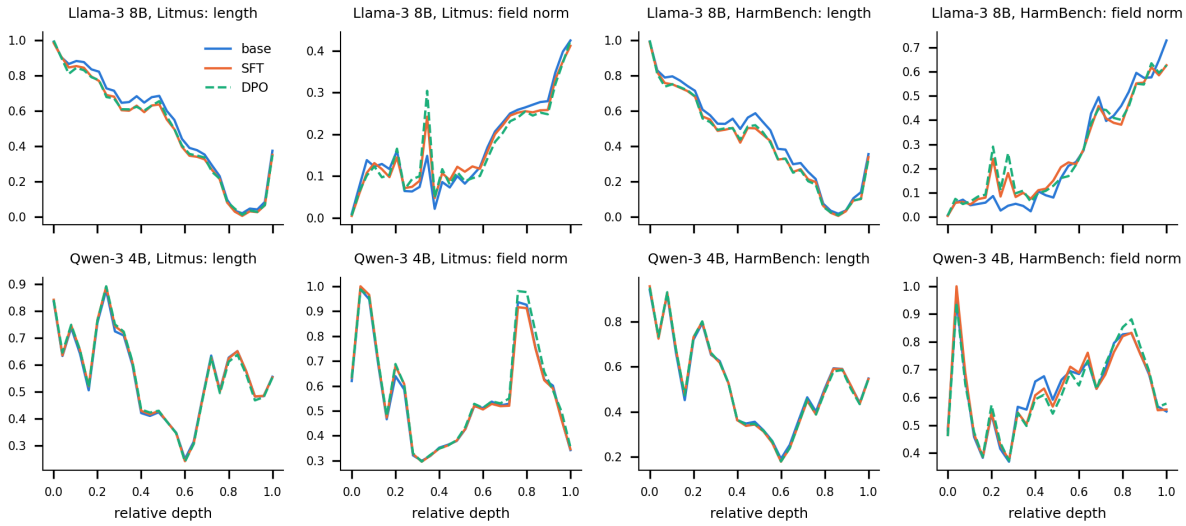


Figure 16: **Alignment stages on Litmus and HarmBench.** Length and field-norm profiles of the base, SFT and DPO versions of Llama-3 8B (top) and Qwen-3 4B (bottom); profiles are rescaled jointly within each probe group, so amplitude differences between stages are kept.

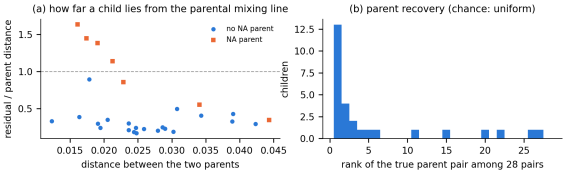


Figure 17: **Merging.** (a) Residual of the best mixture of the true parents against the parents' distance; children with a North America parent are marked. (b) Rank of the true parent pair among all 28 pairs.

P.4 Related-checkpoint retrieval: per-model results

Tab. 45 lists the nearest neighbour of each of the 14 models eligible for lineage retrieval at n=2500. Lineage groups are defined by the base weights a model was trained from: Llama-2 (Llama-2-7B, Vicuna-7B and 13B, Orca-2-7B and 13B), Llama-3 (Llama-3-8B and its instruct version, Tulu-3.1-8B, which is trained from Llama-3.1-8B), Qwen1.5, DeepSeek-LLM and Gemma pairs; Pythia, TinyLlama (Llama-2 architecture, trained from scratch), Phi-2, Mistral and Falcon have no mate in the set. For all 14 models with a mate the nearest neighbour is a mate. Orca-2-13B and Vicuna-13B are mutual nearest neighbours, as are the base and chat models of Qwen1.5, DeepSeek and Gemma. The median distance between models of different lineages is six times that within a lineage (Fig. 12). The groups are defined from documented ancestry, the candidate set is fixed and known, and the set is small and unbalanced, so this is retrieval on a curated set and not provenance verification.

Table 45: **Nearest neighbour for models with a lineage mate** (n=2500, length and field). Singleton models are excluded because they cannot yield a same-lineage retrieval; all 14 eligible models retrieve a mate.

Model	Lineage	Nearest	DTW
Qwen1.5-7B	Qwen-1.5	Qwen1.5-7B-Chat	0.121
Qwen1.5-7B-Chat	Qwen-1.5	Qwen1.5-7B	0.121
Tulu-3.1-8B	Llama-3	Llama-3-8B	0.089
DeepSeek-7B	DeepSeek-LLM	DeepSeek-7B-Chat	0.108
DeepSeek-7B-Chat	DeepSeek-LLM	DeepSeek-7B	0.108
Gemma-7B	Gemma	Gemma-7B-IT	0.258
Gemma-7B-IT	Gemma	Gemma-7B	0.258
Vicuna-13B	Llama-2	Orca-2-13B	0.089
Vicuna-7B	Llama-2	Llama-2-7B	0.064
Llama-2-7B	Llama-2	Vicuna-7B	0.064
Llama-3-8B	Llama-3	Tulu-3.1-8B	0.089
Llama-3-8B-Instruct	Llama-3	Llama-3-8B	0.095
Orca-2-13B	Llama-2	Vicuna-13B	0.089
Orca-2-7B	Llama-2	Vicuna-7B	0.067

P.5 Scope of interpretation

Three observations follow from these cases. First, geometry and behaviour are distinct axes in this set: 4-bit models change perplexity without leaving task variation, and pruned models are far in both. Second, the positive pruning result is a sensitivity-to-destruction result, and the informative next study is an accuracy-preserving pruning or recovery protocol and not a stronger dose. Third, the profile is not a quality score: it measures a particular change in the shape of depthwise predictive computation, and perplexity and task accuracy remain indispensable.

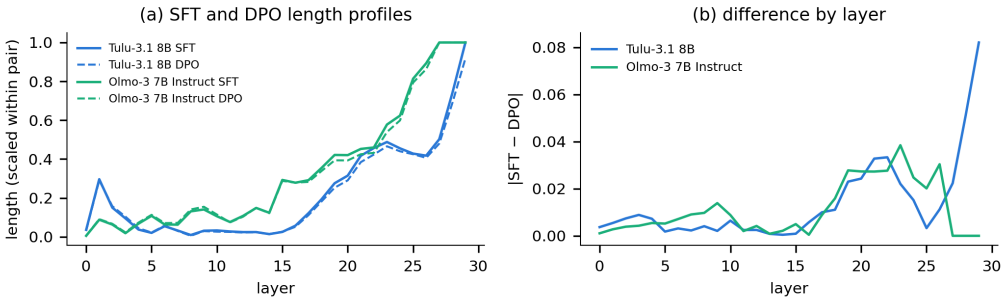


Figure 18: **SFT and DPO length profiles in two published pipelines.** (a) Profiles scaled jointly within each pair (solid SFT, dashed DPO). (b) Absolute difference by layer; for Olmo-3 7B Instruct both checkpoints reach the scaling ceiling at layers 27 to 29.

Table 46: **Lineage retrieval by channel set.** Leave-one-out nearest neighbour shares the lineage (models with a mate), AUC and MRR. Length and field are the main empirical channels; rows labelled "original angle" quantify the contribution of the earlier interior-angle implementation. The last row is computed from the stored report matrices (32 resampled points).

Set	Channels	NN	AUC	MRR
n=2500	$\mathcal{L}$, $\|\tilde{\mathbf{v}}\|$	14/14	0.95	1.00
n=2500	$\mathcal{L}$ only	10/14	0.91	0.84
n=2500	three, original angle	14/14	0.94	1.00
n=1000	$\mathcal{L}$, $\|\tilde{\mathbf{v}}\|$	13/13	0.93	1.00
n=1000	$\mathcal{L}$ only	10/13	0.89	0.85
n=1000	three, original angle	13/13	0.93	1.00
19-model set	$\mathcal{L}$, $\|\tilde{\mathbf{v}}\|$	18/19	0.93	0.96
19-model set	$\mathcal{L}$ only	15/19	0.86	0.85
19-model set	three, original angle	18/19	0.94	0.97
19-model report	three, original angle, 32 points	19/19	0.97	1.00

Reading operations with one measurement exposes null results (alignment stages, light fine-tunes) and, where a task reference has been calibrated on the same probe and scaling, gives a yardstick against which a claimed change can be judged; most case studies here do not yet meet that condition.

Q Recursive self-training with a human-data control

Q.1 Setup

Both arms start from Meta-Llama-3-8B-Instruct (generation 0, measured once and identical in both arms). *Self-training arm:* at each generation the current model answers Alpaca instructions (temperature 0.7, top-p 0.9, at most 128 new tokens), and the next generation is fine-tuned on these answers. *Control arm:* the same schedule, fine-tuned on the human Alpaca answers, so that the amount of training is matched and only the source of the data differs. Training follows the repository driver (learning rate 5×10^{-5}, sequences of 512 tokens, 1000 steps in each of the first three generations and 500 after; the same function sets the schedule in both arms, so both switch at generation 4); the run directories are labelled with 2500 training examples per generation, and the retained metrics files do not record the remaining arguments. The self-training arm has generations 0 to 11, 13 and 14 (the metrics of generation 12 are absent); the control has 0 to 15.

Two measurements were taken at every generation. (i) *Length.* On a fixed set of 512 evaluation texts, Δ_ℓ is the Fisher-Rao step $d(q_{\ell-1}, q_\ell) = 2\arccos\langle\sqrt{q_{\ell-1}}, \sqrt{q_\ell}\rangle$ between consecutive logit-lens readouts, averaged over all valid positions, for the 31 steps between the 32 layers. It is not rescaled, so amplitude is comparable across generations and arms. (ii) *Step speeds.* For 128 prompts truncated to 128 tokens, the original measurement routine records at the last position the 32 steps between the embedding output and the 32 layers as $s_\ell = \|\Pi_{u_\ell}(u_{\ell+1} - u_\ell)\|$ with $u = \sqrt{q}$ and Π_u the projection onto the tangent space at u. Since u_ℓ and $u_{\ell+1}$ are unit vectors, $s_\ell = \sin(d_\ell/2)$ with d_ℓ the Fisher-Rao step, a monotone function of it. The prompts are the same in every generation and arm, so we compare arms by the paired per-prompt difference of the mean speed, self-training minus control, with a 95% percentile interval from 4000 bootstrap resamples of the prompts. The interior-angle score from the same routine is not used in this comparison.

The evaluation set differs between arms from generation 2. Each metrics file records the number of evaluated positions summed over the 31 steps: $2{,}038{,}157 = 31 \times 65{,}747$ in the control and in self-training generations 0 and 1, but $2{,}045{,}721 = 31 \times 65{,}991$ in self-training genera-

tions 2 to 14. The driver explains the difference. In the self-training arm the answer-generation step sets the shared tokenizer to left padding, and from generation 2, when the tokenizer is passed on in memory, the evaluation reuses it. The evaluation shifts inputs and targets by one position and keeps every target whose attention mask is set, so each left-padded text contributes exactly one extra target, predicted from a padding token; the 244 extra positions per step correspond to 244 padded texts. Left padding without explicit position indices also shifts the positions of the real tokens. The length and parameter-effort measurements of the two arms are therefore not computed on identical inputs from generation 2 onwards. If the only change were the 244 added positions, the mean step could move by at most $244/65{,}991 \times \pi = 0.012$, because each step lies in $[0, \pi]$; the position shift has no such bound. The step speeds are computed one prompt at a time, without padding, and are not affected.

Q.2 Results

Mean per-layer length falls in both arms (Tab. 47 and Fig. 19a): from 0.979 to 0.628 at generation 14 under self-training (-36%; Spearman $\rho = -0.987$ with generation), and to 0.660 at generation 14 and 0.652 at generation 15 in the control (-33%; $\rho = -0.997$). The largest single drop is at generation 1 in both arms, and the control drops again at generation 4. The contraction is concentrated early in depth: from generation 0 to the last generation the first, second and last thirds fall by 42%, 39% and 18% under self-training and by 38%, 32% and 24% in the control. The control's decline is measured on identical inputs throughout; the self-training decline crosses the evaluation change between generations 1 and 2. Between the arms the length comparison is confounded: the self-trained profile lies above the control at generation 1 (0.860 against 0.837) and below it from generation 2, exactly when its evaluation inputs change, so the differences at generation 14 (lower in the first two thirds, 0.721 against 0.788 and 0.610 against 0.687, higher in the last, 0.543 against 0.493; Fig. 19b) are not attributed to the data source.

The step speeds give the one between-arm comparison on identical inputs. On the 128 prompts the self-trained model takes larger steps than the control at generation 1 ($+0.015$, interval $[0.008, 0.023]$) and smaller steps at every later matched generation, by 0.008 to 0.096 (Fig. 19c). The interval excludes zero at every generation except generation 3, and from generation 4 onwards between 96 and 125 of the 128 prompts are lower under self-training. These intervals resample prompts, not training runs: with one seed per arm, run-to-run variation of the fine-tuning is not measured and could be as large as the difference. The surfaces of Fig. 20 show the length difference by depth, subject to the evaluation change described above.

Parameter effort. Each run also records, on the same 512 texts, the parameter-space effort $E_\ell = \mathbb{E}_x \|\nabla_{\theta_\ell} \log p(x)\|^2$ of block ℓ, the trace of the observed Fisher information restricted to that block's parameters. Its mean over blocks falls by 80% under self-training (7.0×10^7 to 1.4×10^7; Spearman $\rho = -0.99$ with generation) and by 75% in the control (to 1.8×10^7; $\rho = -0.97$); it is lower under self-training at every generation after the first, a comparison subject to the same evaluation change as length (Fig. 20d). Across generations it moves with the mean length (Spearman 1.00 and 0.98): in the control both stay almost flat over generations 1 to 3 and drop together at generation 4. A smaller E_ℓ means smaller log-likelihood gradients on the evaluation texts, which is what one expects as the model's predictions on them become more concentrated; the coupling with length says that the prediction path shortens as the parameter gradients shrink, without showing which causes which. The ratio Δ_ℓ^2 / E_ℓ, which the repository records as an efficiency, is at most about 10^{-6} and is not interpreted.

The original interior-angle score. The score from the same runs ranges from 11 to 1.8×10^3 under self-training and from 5.7×10^2 to 3.4×10^9 in the control. Values of this size reflect the division by near-zero steps diagnosed in App. F, and they are not used.

Q.3 Scope of interpretation

Repeated fine-tuning on either source contracts the length profile by about a third. Whether self-generated data contracts it further cannot be settled with these runs: the length comparison between arms is confounded by the evaluation change from generation 2, and the unconfounded speed comparison shows a difference in this single run, with one seed per arm and intervals over prompts only. The design isolates the source of the data, but this is one model, one seed and one schedule, and the length and speed measurements use different texts and

positions. No behavioural quantity was recorded in this generation-0–15 profile block, and the field was not logged. A separate matched generation-0–8 evaluation supplies diversity and perplexity outcomes (Tab. 49); the two blocks should not be treated as one jointly sampled experiment.

Q.4 Additional retained recursive trajectories

For these files, a recursive generation is the checkpoint obtained by fine-tuning the preceding checkpoint on responses produced by that preceding checkpoint; a regional label names the corresponding training corpus, not a population. The files retain generation labels and profile arrays but not the optimiser, base-checkpoint revision or training-example manifest, so they preserve outcomes rather than a complete training recipe. The interactive result files retain the numerical arrays passed to Plotly, rather than rasterised images alone. We decode the final `Plotly.newPlot` call in each file, exclude one-point annotation traces, and retain every multi-layer generation trace. The extraction yields 120 checkpoints across nine recursive self-training trajectories (Tab. 48 and Fig. 21). Source paths and SHA-256 hashes, the exact layerwise arrays and the derived summaries are written to `results/collapse_html_trajectories.json` and `results/collapse_html_summary.json` by `extract_collapse_html.py`.

Here the late-layer mean is the arithmetic mean over the eleven plotted decoder layers, 20 through 30. Mean late-layer Fisher–Rao length decreases in every trajectory, by 17.1% to 83.0%. Eight of the nine are strictly decreasing at every retained generation and have Spearman $\rho = -1.00$; linear generation accounts for $R^2 = 0.984$ to 0.996 of their decline. The exception is Latin America, which falls from 0.664 to 0.550 and then plateaus. China falls from 0.692 to 0.217 over generations 0 to 9 (-68.7%), crosses half its initial mean at generation 7, and preserves the shape of its depth profile while its amplitude contracts (first-to-last Pearson $r = 0.989$). These are direct measurements of the plotted prediction profiles. They do not by themselves establish a loss of task performance or output diversity because the chart files retain neither quantity. Their second plotted channel is the original interior-angle score, not corrected spectral curvature; we therefore use Fisher–Rao length for the numerical claim.

Q.5 Matched behavioural outcomes

The generation-0–8 summary pairs the profile with output diversity and held-out perplexity (Tabs. 49 and 50). Under self-training, length falls 31%, distinct-3 falls 30%, self-BLEU rises 57% and WikiText-103 perplexity rises 16%. The human-data control changes by 23%, 4%, 10% and 3%, respectively. These paired trends support a collapse interpretation for this run; they do not define a universal collapse point. An answer-length aggregate supplied with this block is omitted because its recorded unit cannot be reconciled with the 128-new-token generation cap.

Table 47: **Recursive self-training and its human-data control** (Meta-Llama-3-8B-Instruct). Per generation: mean per-layer Fisher-Rao length on the 512 evaluation texts, change from generation 0, and the mean over each third of depth (layers 1 to 11, 12 to 21, 22 to 31), for each arm; paired difference of the mean step speed on 128 prompts (self-training minus control) with its 95% bootstrap interval, and the number of prompts on which self-training is lower. Generation 12 of self-training was not retained.

	self-training					control (human data)					paired, 128 prompts	
gen	mean	change	third 1	third 2	third 3	mean	change	third 1	third 2	third 3	difference [95%]	lower
0	0.979		1.252	0.996	0.660	0.979		1.252	0.996	0.660		
1	0.860	−12%	1.097	0.880	0.580	0.837	−14%	1.079	0.866	0.544	0.015 [0.008, 0.023]	52/128
2	0.802	−18%	1.022	0.810	0.552	0.835	−15%	1.074	0.880	0.526	−0.013 [−0.021, −0.005]	79/128
3	0.774	−21%	0.978	0.786	0.538	0.831	−15%	1.064	0.873	0.532	−0.008 [−0.016, 0.000]	74/128
4	0.725	−26%	0.886	0.729	0.545	0.720	−26%	0.885	0.742	0.517	−0.038 [−0.044, −0.032]	110/128
5	0.712	−27%	0.854	0.706	0.562	0.716	−27%	0.876	0.738	0.517	−0.015 [−0.020, −0.009]	96/128
6	0.704	−28%	0.839	0.704	0.556	0.710	−27%	0.867	0.727	0.520	−0.049 [−0.054, −0.044]	115/128
7	0.703	−28%	0.825	0.690	0.583	0.703	−28%	0.857	0.729	0.509	−0.064 [−0.070, −0.058]	125/128
8	0.684	−30%	0.795	0.665	0.581	0.693	−29%	0.843	0.715	0.506	−0.038 [−0.044, −0.033]	112/128
9	0.669	−32%	0.784	0.645	0.567	0.688	−30%	0.831	0.706	0.513	−0.030 [−0.035, −0.025]	106/128
10	0.682	−30%	0.802	0.660	0.571	0.679	−31%	0.818	0.701	0.505	−0.096 [−0.102, −0.089]	122/128
11	0.670	−32%	0.766	0.646	0.587	0.681	−30%	0.828	0.695	0.505	−0.048 [−0.054, −0.042]	115/128
12						0.669	−32%	0.801	0.680	0.511		
13	0.643	−34%	0.733	0.615	0.573	0.667	−32%	0.791	0.693	0.506	−0.082 [−0.088, −0.076]	124/128
14	0.628	−36%	0.721	0.610	0.543	0.660	−33%	0.788	0.687	0.493	−0.058 [−0.063, −0.052]	121/128
15						0.652	−33%	0.771	0.674	0.501		

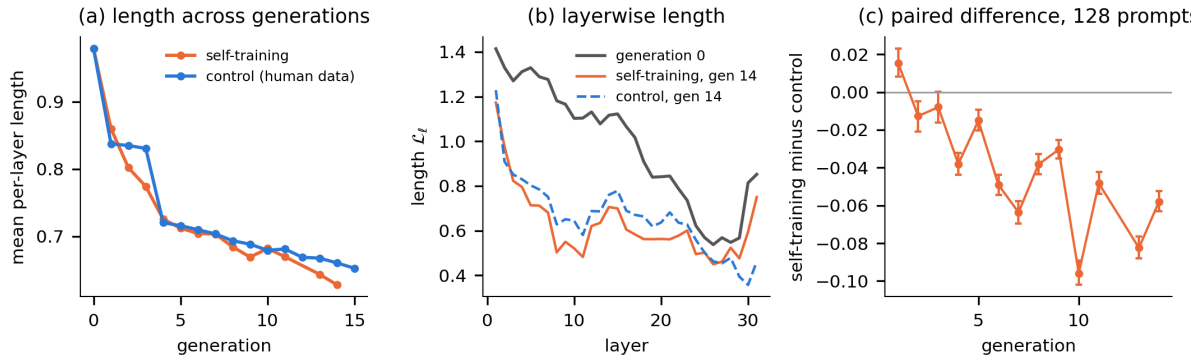


Figure 19: **Recursive self-training against a human-data control.** (a) Mean per-layer length on the 512 evaluation texts per generation. (b) Layerwise length at generation 0 and at generation 14 of each arm. (c) Paired difference of the mean step speed on 128 prompts, self-training minus control, with 95% bootstrap intervals; below zero, the self-trained model moves less.

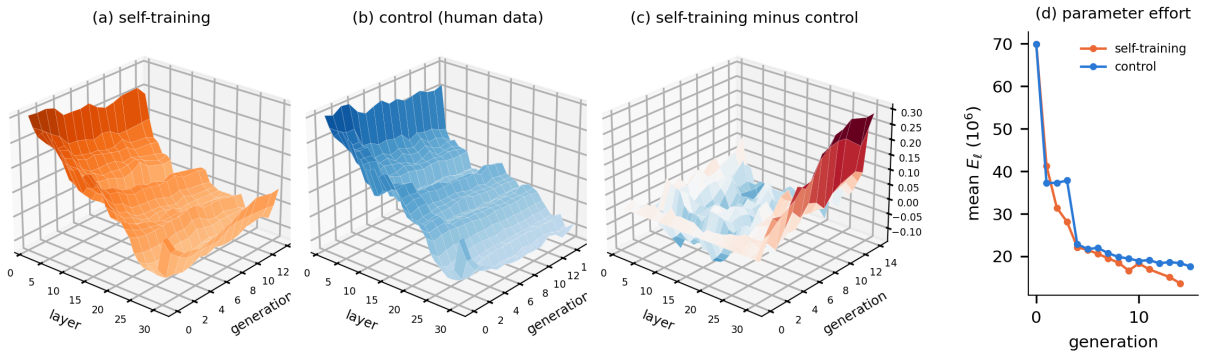


Figure 20: **Recursive runs as surfaces.** (a, b) Per-step length over layer and generation for each arm (no rescaling; generation 12 of self-training is interpolated by the surface). (c) Self-training minus control at the 14 shared generations: blue where the self-trained model moves less; from generation 2 the self-training evaluation used left padding, so this panel is not a clean comparison. (d) Mean parameter effort E_ℓ per generation.

Table 48: **Recursive self-training trajectories decoded from retained Plotly arrays.** "Half" is the first generation whose mean late-layer length is no more than half its generation-0 value. The dash means that the trajectory does not reach half. R^2 is from a linear fit of mean length on generation.

Trajectory	generations	checkpoints	initial	final	change	half	ρ / R^2
Llama	0–10	11	0.601	0.146	−75.7%	6	−1.00 / 0.986
Africa	0–15	16	0.699	0.172	−75.4%	10	−1.00 / 0.995
Asia	0–11	12	0.693	0.118	−83.0%	7	−1.00 / 0.996
Australia	0–16	17	0.506	0.245	−51.5%	16	−1.00 / 0.984
China	0–9	10	0.692	0.217	−68.7%	7	−1.00 / 0.996
Europe	0–13	14	0.495	0.236	−52.2%	13	−1.00 / 0.991
Latin America	0–13	14	0.664	0.550	−17.1%	–	−0.80 / 0.441
Middle East	0–9	10	0.706	0.227	−67.9%	7	−1.00 / 0.996
North America	0–15	16	0.499	0.223	−55.3%	13	−1.00 / 0.990

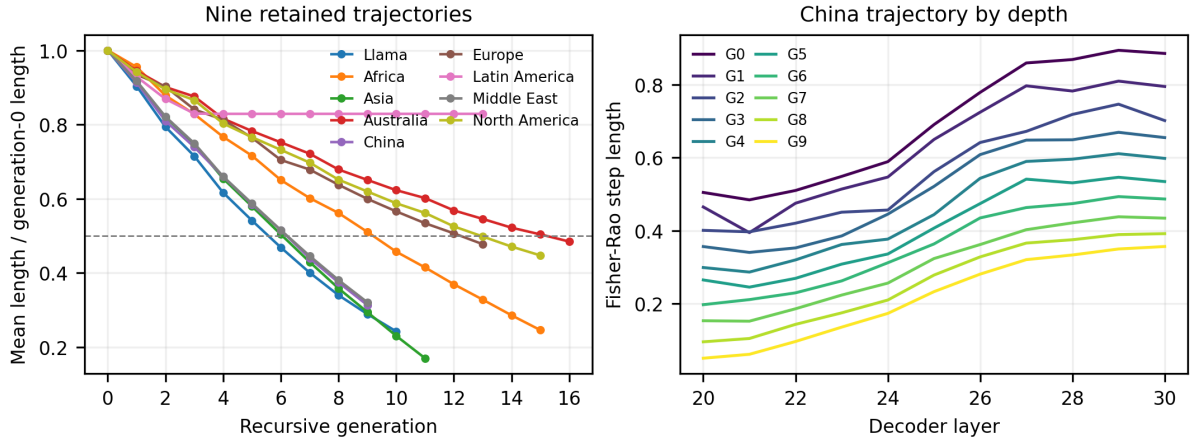


Figure 21: **Additional recursive self-training trajectories.** Left: mean late-layer Fisher–Rao length relative to generation 0 for all nine retained trajectories; the dashed line marks one half. Right: the unscaled China length profile across decoder layers for generations 0 to 9. The figure is regenerated directly from the Plotly arrays; no corrected-curvature values are inferred from the original interior-angle axis.

Table 49: **Recursive self-training and human-data control.** Distance is the retained aligned mean-squared cost from generation 0; distinct-3 and Alpaca perplexity use the matched evaluation protocol.

	self-training				human-data control			
gen.	length	cost_0	distinct-3	Alpaca PPL	length	cost_0	distinct-3	Alpaca PPL
0	.912	.000	.711	5.83	.912	.000	.711	5.83
1	.858	.013	.699	5.88	.875	.007	.709	5.84
2	.823	.026	.681	5.95	.846	.012	.706	5.87
3	.786	.040	.653	6.03	.819	.017	.703	5.90
4	.748	.056	.621	6.15	.794	.022	.699	5.94
5	.713	.070	.591	6.31	.773	.027	.695	5.99
6	.681	.083	.560	6.54	.748	.031	.691	6.04
7	.652	.095	.529	6.82	.724	.035	.688	6.09
8	.631	.108	.501	7.15	.702	.039	.684	6.15

Table 50: **Additional recursive endpoints** with documented units.

Metric	gen. 0	self gen. 8	control gen. 8
distinct-1	.286	.238	.278
distinct-2	.548	.421	.530
distinct-3	.711	.501	.684
self-BLEU ↓	.326	.512	.359
WikiText-103 PPL	7.12	8.23	7.34
projected field norm	.318	.261	.301

S Comparison with other readouts

Most comparison methods answer a different scientific question from nDNA. Rather than assigning contestable yes/no properties to each method, Tab. 51 maps the reader's question to the appropriate tool and states the narrower role of nDNA.

S.1 What we add

The Fisher-Rao line element has an intrinsic scale on a fixed categorical support: it does not inherit arbitrary hidden-state units. This is weaker than a universal scale across language models: tokenisers define different categorical spaces, temperature changes both the path and the belief push, and the DTW experiments use blockwise normalisation. Canonical scale is therefore marked partial. The combination that is new is being coordinate-free, retaining a condition-specific direction, and needing no auxiliary training, together with an additive depth quantity that is calibrated against task variation.

S.2 Qualitative comparison by scientific question

Logit, tuned and future lenses. These methods answer what an intermediate state already predicts. nDNA takes that readout as its starting point and asks a second-order question: how the sequence of predictions moves and turns across depth. The ordinary logit lens is cheap but can misread early layers; the tuned lens corrects this by learning a translator per layer, at the cost of auxiliary training and a new source of cross-model variation (Belrose et al., 2023). Future-lens methods give richer token-level forecasts and are preferable when the target is the content available at one layer (Pal et al., 2023; Ghandeharioun et al., 2024).

Adjacent-layer KL and entropy. This is the closest scalar baseline. KL localises predictive change and is inexpensive but is asymmetric and violates the triangle inequality; entropy records confidence, not movement, and two distributions with equal entropy can have disjoint mass. Fisher-Rao length is a symmetric metric, curvature adds directional revision and the field adds direction. KL remains a useful robustness check because it induces the same quadratic form locally (Thm. 1(iii)); disagreement at large steps diagnoses a finite-distance choice, not an error.

CKA and SVCCA. Representation similarity compares activation matrices after choosing examples, layers and alignment conventions, and is a strong tool when asking whether two hidden spaces encode similar variation, including across vocabularies (Kornblith et al., 2019; Raghu et al., 2017; Klabunde et al., 2025). It does not give an additive probability-space path length and depends on sampling and preprocessing; nDNA trades breadth for a direct prediction-space readout.

Probes and sparse autoencoders. They can name content that nDNA cannot, but their answer depends on a learned auxiliary model and a target concept, whereas nDNA needs only the next-token labels the model was trained on (Cunningham et al., 2023). They are follow-up tools after a descriptive depth difference is observed; the current patching result does not validate that difference as causal localisation.

Attention rollout and flow. They expose token-to-token routing but are complicated by residual pathways, head aggregation and the fact that attention is not itself a causal effect (Abnar and Zuidema, 2020; Jain and Wallace, 2019; Wiegreffe and Pinter, 2019). nDNA measures predictions after the whole residual update and avoids choosing a head-compression rule, but cannot replace routing analysis.

Activation patching and circuits. Interventions answer whether replacing a component changes the output and are the right standard for causal claims, but they need many counterfactual passes and depend on the corruption and the patching site (Heimersheim and Nanda, 2024; Conmy et al., 2023). nDNA is observational and cheaper; the non-significant depth-controlled correlations (Tab. 20) show that it is not presently a proxy for causal effect.

Weight-space views. Task vectors and deltas are natural when an operation is additive in parameters, but they cannot compare architectures with incompatible shapes and can assign large distance to functionally redundant reparametrisations (Ilharco et al., 2023; Ainsworth et al., 2023). nDNA compares realised predictive trajectories but needs white-box access and a compatible output space.

Lineage methods. Output-based phylogenies and weight fingerprints answer the lineage question without profiling depth (Yax et al., 2025; Zeng

Table 51: **Which tool answers which question.**

Question	Best-suited tool	What it needs	Role of nDNA
What does a layer already predict?	logit, tuned or future lens	output head; trained translator for the tuned lens	starting point of the readout
Where does the selected prediction readout differ across depth?	nDNA	weights, one or two forward passes	descriptive profile comparison
Which component causes the output?	activation patching, ACDC, causal scrubbing	many counterfactual passes	cannot replace; may prioritise descriptive depth bands
Which feature or concept is represented?	probes, sparse autoencoders	trained auxiliary model	follow-up to a descriptive profile difference
Are two hidden spaces similar?	CKA, SVCCA	activations of both models	different object
Which tokens route information?	attention rollout and flow	attention tensors	different object
How did the parameters change?	task vectors, weight deltas	matching architectures	different object
Is a model usable?	perplexity, accuracy	benchmark	required companion; never alone
Is a checkpoint a descendant of a known base?	nDNA lineage retrieval; output phylogeny; weight fingerprints	white-box profile; outputs; weights	retrieval on curated cohorts; not compared head to head

et al., 2024). nDNA retrieves related checkpoints as a by-product of a depth-resolved profile; we do not claim to beat these on lineage and have not compared head to head. The current cohorts also do not separate profile information from tokenizer, depth and architecture metadata.

Perplexity and accuracy. They remain indispensable for assessing whether a modified model retains the evaluated behaviour and prevent catastrophic pruning from being read as an interesting signature. A layerwise profile can show where its selected readout differs, but this paper does not validate that location causally or show a matched-performance case in which it improves a decision. Every operation result should report behaviour and nDNA together.

S.3 What we lack

We do not name what changed: that is for features and circuits. **We do not establish effect:** the profile is observational. **Readout dependence remains:** the tuned-lens check in Tab. 27 preserves the main decoder result on one subset, but does not cover every architecture or operation. **We pool over positions:** no token-level attribution. **We need weights:** App. U.

S.4 Empirical comparison

The matched comparison in Tab. 26 evaluates Fisher–Rao length against forward KL, Jensen–Shannon divergence, entropy and corrected curvature on the same association, identification, retrieval and threshold decisions. Fisher–Rao length leads each reported outcome. CKA and weight-space similarity are not available on the same candidate set, so the comparison does not establish superiority over representation- or parameter-space methods.

T Why the channels are stacked rather than combined

The channels could be combined in two places: within a layer, into one number per layer, or over the whole profile, into one number per checkpoint. The natural candidates are a product (the repository's composite score is $\sum_\ell \omega_\ell \kappa_\ell \mathcal{L}_\ell \|\mathbf{v}_\ell\|$ with fixed weights), a sum, a difference, or a sum of logarithms. The protocol uses none of them: the channels enter DTW as separate coordinates of $\mathbf{z}_i$ (Eq. (7)), and the cost $\|\mathbf{z}_i - \mathbf{z}'_j\|^2 = \sum_c (z_{i,c} - z'_{j,c})^2$ adds their squared differences. This appendix gives the reasons, one general and three specific to the operators, and tests them.

No per-layer number keeps both channels.

Proposition 9. *Let $U \subset \mathbb{R}^2$ be open and $f : U \to \mathbb{R}$ continuous. Then f is not injective.*

Proof. Suppose f is injective. Take a circle $S \subset U$. $f(S)$ is compact and connected, hence an interval $[m, M]$, and $m < M$ because f is injective and S has more than one point. Choose $s_0 \in S$ with $f(s_0) \in (m, M)$. $S \setminus \{s_0\}$ is connected, so $f(S \setminus$

$\{s_0\}$) is connected; by injectivity it equals $[m, M]\setminus\{f(s_0)\}$, which is not connected. Contradiction. □

Any continuous rule that turns $(\mathcal{L}_\ell, \|\mathbf{v}_\ell\|)$ into one number therefore assigns the same value to different pairs: the product to every pair on a hyperbola $\mathcal{L}\|\mathbf{v}\| = c$, the sum to every pair on a line $\mathcal{L} + \|\mathbf{v}\| = c$, the difference to every pair on a line $\mathcal{L} - \|\mathbf{v}\| = c$. Two checkpoints that differ by moving along such a set are indistinguishable after combination. Stacking is injective: the cost is zero at aligned layers only if every channel agrees.

The product gates. Since $\partial(\mathcal{L}\|\mathbf{v}\|)/\partial\mathcal{L} = \|\mathbf{v}\|$, a change in length is invisible wherever the field norm is small, and conversely; one channel near zero silences the other. Under the affine rescaling the dashboards apply to each channel, $(\alpha\mathcal{L} + \beta)(\gamma\|\mathbf{v}\| + \delta) = \alpha\gamma\mathcal{L}\|\mathbf{v}\| + \alpha\delta\mathcal{L} + \beta\gamma\|\mathbf{v}\| + \beta\delta$, so the product of scaled channels is not a function of the product of the raw channels: it mixes in additive terms whose weights are set by the arbitrary floor of the scaling. The logarithm of a product, $\log\mathcal{L} + \log\|\mathbf{v}\|$, turns the product into a sum but is undefined at zero and weights small values by $1/\|\mathbf{v}\|$, so it is dominated by the least reliable readings.

Sums and differences need a common unit. Length is a Fisher-Rao arc in radians, about 1 per layer; the field norm is a tangent vector per unit logit step, about 10^{-3} in the unscaled exports (App. L). A raw sum is length, and a raw difference is length with a small offset. After rescaling, a sum or difference weights the channels equally by convention, and a rise in one channel can be cancelled by a fall in the other, so a change that moves only the field (as alignment might) can disappear. A difference also has no meaning in sign: $\mathcal{L} - \|\mathbf{v}\|$ compares quantities of different kinds.

What stacking does and does not assume. Stacking also needs a common scale, which the protocol supplies by a stated rescaling, and it also weights the rescaled channels equally. Its advantages are that no channel gates or cancels another, that the cost is additive over channels so each channel's contribution can be reported separately, as in Tab. 2, and that it loses nothing at a layer.

Test on the retained profiles. Tab. 52 compares stacking with each per-layer combination (DTW on the combined channel, rescaled per profile) and with whole-profile scalars (absolute difference of the scalars), on identification across sample size and on lineage retrieval at n=2500. Every per-layer combination does worse than stacking on both (10 to 13 of 18 identified and 10 or 11 of 14 retrieved, against 14 and 14); length alone identifies best (18 of 18) but retrieves worse (10 of 14), so stacking is the only form that keeps the field's contribution to retrieval. Whole-profile scalars identify 1 to 4 of 18 and retrieve 0 to 9 of 14, although their rank order across sample sizes is stable (Spearman 0.82 to 0.95): they order checkpoints consistently but cannot tell them apart. The choice of operator matters less than keeping the depth axis and the channels separate.

Table 52: **Stacking against combining.** Identification of 18 checkpoints across sample size (n=1000 against n=2500), lineage retrieval at n=2500 (nearest neighbour is a relative; AUC), and, for whole-profile scalars, the Spearman correlation of the scalar between the two sample sizes. Per-layer combinations are compared with DTW; scalars by absolute difference.

	Combination	ident.	retr.	AUC	ρ
per layer	stacked $(\mathcal{L}_\ell, \|\mathbf{v}_\ell\|)$ (protocol)	14/18	14/14	0.95	
	$\mathcal{L}_\ell$ alone	18/18	10/14	0.91	
	product $\mathcal{L}_\ell\|\mathbf{v}_\ell\|$	13/18	11/14	0.85	
	geometric mean	11/18	11/14	0.88	
	sum $\mathcal{L}_\ell + \|\mathbf{v}_\ell\|$	10/18	11/14	0.83	
	difference $\mathcal{L}_\ell - \|\mathbf{v}_\ell\|$	10/18	10/14	0.90	
whole profile	dashboard composite (product of three)	4/18	6/14	0.64	0.95
	$\sum_\ell \mathcal{L}_\ell\|\mathbf{v}_\ell\|$	2/18	9/14	0.65	0.89
	$\sum_\ell(\log\mathcal{L}_\ell + \log\|\mathbf{v}_\ell\|)$	1/18	8/14	0.87	0.90
	$\sum_\ell(\mathcal{L}_\ell + \|\mathbf{v}_\ell\|)$	3/18	9/14	0.58	0.95
	$\bar{\mathcal{L}} - \overline{\|\mathbf{v}\|}$	3/18	0/14	0.79	0.82

The composite score of the dashboards. In the archived dashboards, 65 of 190 profiles are exactly zero (34%), with whole groups at zero (alignment, distillation, HarmBench and RLHF), and the nonzero values span 0.3 to 10^{16}, including 8.3×10^6 for one probe of Llama-3 8B. On the same models the scalar shrinks by 1.3 to 2.9 times between n=1000 and n=2500 (App. N). This particular product, with this range, collapsing to zero for a third of the profiles and depending on n, cannot support comparison between operations, and Tab. 52 shows that the failure to separate checkpoints is shared by every whole-profile scalar we tried. Its rank order across models is stable (Spearman 0.95 between the two sample sizes), so it carries some information but is not a calibrated number.

U Scope, limitations and constructions not adopted

U.1 Models we cannot measure

nDNA needs the hidden state at every layer and the output head. An API model returns the final layer only, often truncated to the top few tokens, so the depth axis is not exposed and no prompting strategy recovers it. This is a property of the object, not a defect: every method that reads internal computation (patching, sparse autoencoders, circuits) has the same restriction. Top-k log-probabilities allow final-distribution comparisons but not a path. The one indirect route is to profile open-weight models that are claimed to be replicas or derivatives of a closed model.

U.2 Full list of limitations

1. **Observational, not interventional.** The profile shows that computation changed at a depth, not that changing it would change behaviour (Tab. 20).
2. **No feature-level or validated localisation account.** We report where the selected readout differs across depth; we do not identify a feature, head or circuit, and patching does not validate those bands causally.
3. **Readout reliability in early layers** is known to be weak in some models; a trained-readout robustness run is missing.
4. **Decoder-only transformers only.** Encoder-only and encoder-decoder models have different structure and are untested.
5. **Depth and vocabulary heterogeneity.** DTW will align profiles of different lengths, but the result is meaningful only if matched layers play comparable roles; shape comparisons are restricted to depth ratios below two. Arc length lives on a sphere whose dimension is the vocabulary size, so cross-tokeniser length comparisons need normalisation.
6. **Stability is empirical and descriptive.** Ten tasks, four decoders, one seed, no prompt-resampling intervals; the 45 task pairs share ten tasks and are not independent; no bound on tolerated distribution shift, and degradation is expected for unusual inputs (very long documents, adversarial prompts, code in a model with little exposure).
7. **Condition specification.** The field depends on the prompt distribution chosen to represent a condition; different sets for one nominal condition can give different fields.
8. **Position pooling.** We average over supervised positions, so no token-level attribution.
9. **Estimator versions.** Corrected exterior-turn curvature and exact parallel transport are defined and tested mathematically. Corrected curvature is also evaluated in the matched baseline block, but the default comparator uses length and the projected-mean field norm; the exact-transport approximation error remains unmeasured.
10. **Position pooling.** The dashboards use all valid positions for length and the last eight for the field by design. A last-eight matched analysis preserves the channel associations, but no token-level attribution is attempted.
11. **Retrieval confounds.** Related-checkpoint retrieval is matched by architecture and tokenizer metadata in these cohorts. It is evidence of association in a fixed candidate set, not independent lineage information or provenance verification.
12. **Operation calibration.** The frozen task-only analysis confirms the quantisation, pruning and sibling decisions in four decoders, but it remains tied to the same ten-task suite and one seed.
13. **Recursive outcomes.** One matched run links self-training to path contraction, reduced diversity and higher perplexity relative to a human-data control. The additional HTML trajectories do not retain behavioural metrics or complete training manifests, so they cannot establish a universal collapse point.
14. **Cost of the final-output field.** The layer-local field needs no backward pass; the alternative final-output sensitivity does, and is the dominant cost on large models.
15. **Limited mapping to downstream outcomes.** Compression and recursive training include aggregate behavioural outcomes, but alignment, merging and distillation remain profile-only; no release criterion is validated.
16. **Dashboard scales are local.** Per-profile scaling retains shape only; groupwise scaling re-

tains relative amplitude within its group. The task threshold and behaviour block include the original interior-angle score.

17. **Incomplete run metadata.** Runtime hardware, peak memory, all Hub revision hashes and the ten-task dashboard sample sizes are not in the retained manifests; reported times are therefore not hardware-normalised benchmarks.

18. **Closed models are out of scope.**

U.3 Constructions we tried and did not adopt

The same three quantities can be defined on other parts of a transformer. We built four such variants before settling on predictions; each is mechanistically reasonable and each failed a requirement that the final method must meet: extractable without special instrumentation, shape-stable under prompt variation, and independent of tuning knobs. **Attention as transport:** treat each layer's attention as a stochastic matrix moving mass between positions, aggregate heads by an entropy-based weighting, and read length from a Fisher-type metric on the kernels. It needs full attention tensors, which scale with the square of the sequence length and are often unavailable, and the head-aggregation weights are a free choice that changes shape agreement. **Parallel transport on hidden states:** align consecutive layers by orthogonal Procrustes, whiten by the layer covariance, take covariant differences. This is the principled way to measure the hidden path and the variant we recommend for hidden-space quantities, but it costs a covariance and a decomposition per layer and the ridge parameter changes both curvature and shape. **Feed-forward Jacobian geometry:** length from the motion of the log singular values of the expected block Jacobian. This is an operator deformation measure, not a distance between predictions, and estimating it robustly needs heavy linear algebra or approximations with their own parameters. **Key-value stream as a flow:** fit a linear map carrying the key matrix from one layer to the next, split it into rotation and stretch. The fit is frequently ill-conditioned because key matrices are token-dependent and low rank; matrix logarithms amplify the noise and regularisation choices break shape stability. These trials motivated our use of prediction-space readouts: they avoid several alignment and storage choices required by the alternatives above. They do not establish that prediction profiles are universally stable or hyperparameter-free; the readout, temperature, aggregation and profile scaling remain part of the measurement.